RWTH Aachen University

Interpretable Verification of Coordinated Sybil Wallets in Blockchain Airdrops via On-Chain Behavioral Analysis

Bachelor Thesis

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Declaration of AI Assistance

In the preparation of this thesis, generative artificial intelligence (AI) tools were used exclusively for language refinement and proofreading purposes. All scientific content, ideas, analysis, and conclusions were developed independently by the author.

The use of AI tools did not influence the research design, methodology, results, or interpretation of findings.

Aachen, August 20, 2026 Zhixiong Wang

Abstract

Blockchain airdrop campaigns are increasingly vulnerable to Sybil attacks, where a single entity controls multiple wallets to obtain unfair rewards. While existing detection methods achieve strong performance, they often lack interpretability and fail to provide verifiable evidence linking coordinated wallets.

In this thesis, we identify a limitation of traditional similarity modeling paradigms in this domain. Existing approaches primarily rely on either magnitude-based metrics (e.g., Euclidean distance) or angle-based metrics (e.g., cosine similarity), both of which can be insufficient for capturing fine-grained temporal coordination in highly synchronized Sybil behaviors.

To address this limitation, we propose an interpretable verification framework based on on-chain behavioral analysis. The framework uses a weighted decay mechanism that integrates feature-level weighting with temporal decay, enabling joint modeling of behavioral intensity and coordination patterns while preserving feature-level interpretability. Combined with funding graph decomposition and DBSCAN clustering, the framework identifies behaviorally coherent wallet subgroups within coordinated Sybil wallets.

To further handle the inherent noise produced by density-based clustering, we introduce a contamination inference framework that re-evaluates unassigned nodes using graph-based exposure and behavioral similarity to known Sybil structures. This enables additional evidence to be extracted from noise nodes that are not captured by the primary clustering stage alone.

We evaluate our approach on real-world blockchain datasets from three blockchain airdrop campaigns, examining its ability to identify coordinated Sybil patterns and provide interpretable behavioral and funding evidence. The results demonstrate that the framework can provide high-confidence and interpretable verification of strongly coordinated Sybil activity, while extending analysis to previously unresolved noise nodes.

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Chapter 1

Introduction

1.1 Background

Blockchain airdrop campaigns are widely used to distribute tokens and incentivize user adoption. These campaigns typically reward participants based on on-chain activities, such as interacting with decentralized applications or conducting transactions with specific tokens.

However, this mechanism is vulnerable to *Sybil attacks*, where a single entity controls multiple wallets to obtain a disproportionate share of rewards. Such behavior undermines the fairness of token distribution and distorts key metrics used to evaluate user engagement. As a result, these attacks can erode community trust and lead to financial losses for project teams.

Detecting Sybil wallets has therefore become a critical task in maintaining the integrity of airdrop campaigns. While machine learning approaches have shown strong performance in identifying suspicious wallets, they often operate as black-box models and provide limited insight into the reasoning behind their predictions.

To address this limitation, this thesis proposes a complementary verification framework that leverages graph-based and behavioral analysis. The goal is not only to validate the experiment results produced by machine learning models but also to provide interpretable evidence that explains why certain wallets are likely controlled by the same entity.

1.2 Motivation

Recent advances in machine learning, including gradient boosting, clustering techniques, and graph neural networks, have enabled effective detection of Sybil accounts. These methods typically output wallet-level predictions in the form of probability scores or binary classifications.

Despite their effectiveness, these approaches suffer from limited interpretability (See Related Work Section). In the context of airdrop campaigns, simple classification results are often insufficient. Project teams require clear and explainable evidence that demonstrates relationships between flagged wallets, particularly when decisions may affect user rewards or trigger disputes.

Moreover, many existing methods focus on detection accuracy without explicitly revealing the underlying connections between wallets. As a result, they fail to capture the structural and behavioral patterns that link multiple wallets to a single controlling entity. This lack of

transparency makes it difficult to justify decisions and handle potential appeals from users.

Motivated by these challenges, this thesis proposes a complementary approach that utilizes on-chain transaction graphs and behavioral patterns to provide interpretable insights. By uncovering relationships between wallets and highlighting shared behaviors, the proposed framework aims to enhance both the transparency and reliability of Sybil detection in blockchain airdrop systems.

1.3 Scope

This approach focuses on highly coordinated, automated Sybil wallets that exhibit star-like or tree-like interaction patterns, which represent some of the most prevalent and structurally identifiable attack forms in airdrop campaigns. By concentrating on these well-defined patterns, the framework enables robust and interpretable analysis of coordinated behavior.

While more adaptive or less structured Sybil strategies fall outside the primary focus, the proposed solution concept adopts an onchain-data-driven perspective to generate explainable evidence, effectively complementing existing machine learning-based detection methods in the current scope.

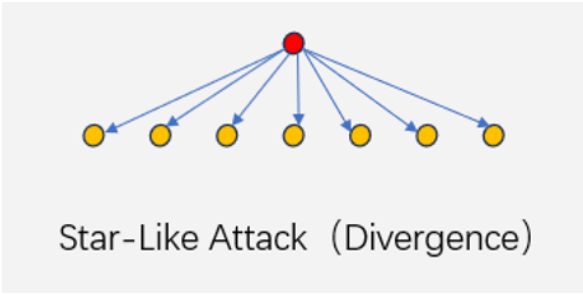


Figure1: Star-like Sybil Structure

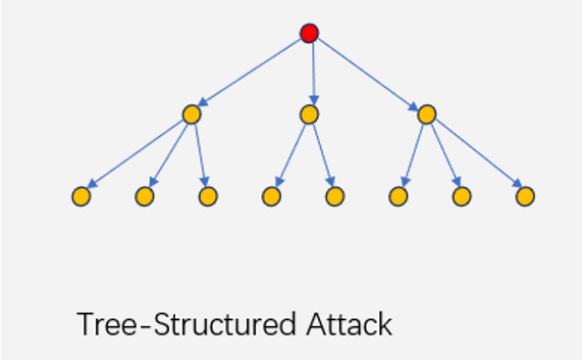


Figure2: Tree-like Sybil Structure

1.4 Research Questions

The goal of this thesis is to design an interpretable hybrid framework that verifies Sybil detection results using multiple independent evidence signals.

The research questions of this thesis are structured around four dimensions: (i) a systematic comparison of existing Sybil detection methods and their limitations, (ii) modeling of Sybil activity (iii) the design of effective features and similarity measures for implementation, and (iv) an analysis of robustness and potential adversarial weaknesses of the proposed approach.

- How do existing sybil detection methods perform, and what are their strengths and key limitations?
- To what extent can Sybil behavior be modeled from on-chain wallet data, and what criteria can be used to identify potential sybil activity?

- Which features and wallet similarity measures are effective for capturing entity-level coordination among wallets?
- What are the limitations and potential vulnerabilities of the proposed method, and how can adaptive adversaries exploit its assumptions to evade detection?

Chapter 2

Related Work

Detecting Sybil attacks has been an important research topic across distributed systems, social networks, and more recently blockchain ecosystems. Existing research addresses this problem from very different angles, involving resource or metadata signals, interaction behavior, graph structure, and hybrid representation learning methods. In this chapter, we group representative works by method category and compare them along four dimensions: detection mechanism, interpretability, performance, and limitations, summarized in Table 2.5. Within each category, we discuss representative approaches and their limitations, then relate these limitations to the design choices adopted in this thesis.

2.1 Graph Structure-Based Sybil Inference

Graph-structure-based methods treat the relationships between nodes as the primary source of evidence for Sybil detection. Early defenses such as SybilGuard and SybilLimit are built around the assumption that honest users form a well-connected region in the social graph, while Sybil identities are connected to the honest region through a relatively small number of attack edges [13, 14]. SybilGuard exploits this structural property through random routes over the social graph, while SybilLimit improves the efficiency and theoretical guarantees of this approach through random walks and short-cycle intersections. These methods established a fundamental principle for graph-based Sybil detection: the connectivity structure of a network can provide useful evidence for distinguishing Sybil identities from honest users even when individual node attributes are unavailable.

SybilInfer subsequently shifted graph-based detection from structural membership tests toward probabilistic inference [1]. Instead of directly determining whether a node belongs to an honest region, SybilInfer models the probability that nodes belong to honest or Sybil regions and applies Bayesian inference over the observed social graph. This probabilistic perspective was further developed by SybilBelief, which formulates Sybil detection as a semi-supervised inference problem and propagates information from a small set of known honest and Sybil nodes through the network using a Markov random field and belief propagation [4]. The resulting Sybil likelihoods therefore depend not only on individual node properties but also on the relational context in which the nodes are embedded.

SybilFrame further extends structure-based inference by incorporating prior knowledge

about both nodes and edges within a defense-in-depth framework [3]. Rather than relying on a single structural signal or a strict trust assumption, SybilFrame combines multiple sources of prior information to improve robustness against variations in network structure and attack conditions. This progression from SybilGuard and SybilLimit through SybilInfer and SybilBelief to SybilFrame reflects an increasing sophistication in graph-based inference: random-walk-based structural separation is gradually augmented with probabilistic reasoning, semi-supervised propagation, and more flexible structural priors.

More recently, SybilGAT extends this line of work using graph neural networks and attention-based message aggregation [5]. Instead of explicitly specifying how information should propagate between neighboring nodes, SybilGAT learns representations from graph structure and assigns different attention weights to neighboring nodes during aggregation. This allows the model to learn which structural relationships are more informative for Sybil classification and provides greater flexibility than manually specified propagation rules. In this sense, SybilGAT represents a transition from explicitly designed probabilistic graph inference toward learned graph representations, while retaining relational structure as the primary source of evidence.

Despite this methodological progression, the classical graph-based Sybil literature was predominantly developed for social-network or decentralized/P2P settings, where edges encode trust, friendship, or other socially meaningful relationships. This assumption does not directly transfer to blockchain transaction graphs. A blockchain edge represents an economic transaction rather than a trust relationship, and multiple transactions between wallets do not necessarily indicate that the wallets belong to the same entity. Conversely, Sybil actors may deliberately create additional transactions or funding links to alter their structural position in the graph. As a result, graph proximity, belief propagation, or learned attention weights can provide evidence of suspicious relational structure, but they do not by themselves establish the behavioral or economic mechanism connecting multiple wallets to a common controller.

Therefore, graph structure provides a powerful foundation for identifying coordinated or anomalous regions, but structural inference alone is insufficient for explaining why a group of blockchain addresses should be treated as a coherent Sybil cluster. In particular, the learned representations and probabilistic scores produced by these methods provide relational evidence rather than direct transaction-level explanations. This limitation motivates hybrid approaches that combine graph structure with explicit behavioral, temporal, transactional, or metadata-based features, allowing detection models to capture both relational context and the underlying activity of blockchain addresses.

2.2 Hybrid and Representation Learning Approaches

Hybrid methods combine graph structure, behavioral features, and machine learning to capture complementary aspects of Sybil and airdrop-hunter behavior.In blockchain airdrops, Liu et al. propose a subgraph-based feature propagation and fusion approach for Sybil address detection [8]. The method constructs multi-layer transaction subgraphs around each address and combines lifecycle-aware temporal features, transaction-amount features, and network-structural features extracted at different depths. These heterogeneous features are propagated and fused before being fed into a LightGBM classifier. Evaluated on 193,701 addresses, including 23,240

Sybil addresses, the approach achieves precision, recall, F1-score, and AUC values above 0.90, demonstrating the effectiveness of combining transaction topology with behavioral and lifecycle information for Sybil detection [8].

A more explicit example of heterogeneous graph learning is ARTEMIS, proposed by Zhou et al. for airdrop-hunter detection in NFT markets [15]. ARTEMIS combines multimodal NFT information with transaction-graph representations and domain-specific behavioral features. Specifically, it extracts semantic information from NFT images and textual metadata using pretrained Transformer models, aggregates information from transaction neighbors through multi-hop graph aggregation, and incorporates manually designed features motivated by airdrop-hunter behavior. Evaluated on transaction data from the Blur NFT marketplace, ARTEMIS achieves a precision of 0.820, recall of 0.833, and F1-score of 0.826, obtaining the highest precision and F1-score among the evaluated methods, including conventional machine-learning models and graph neural networks. In particular, its F1-score exceeds those of GCN (0.752), GraphSAGE (0.701), GAT (0.579), and GIN (0.776) [15]. These results demonstrate that combining relational transaction information with domain-specific and multimodal NFT features can improve the ability to distinguish airdrop hunters from legitimate active traders.

However, the predictive strength of such supervised and representation-learning approaches does not necessarily translate into interpretable or transferable detection decisions. Recent work by Li et al. identifies three broader limitations in supervised airdrop-hunter detection: inconsistent definitions and labeling, weak explainability, and poor cross-context generalization [7]. In particular, the authors note that supervised methods commonly face a "black-box" challenge, making their decisions difficult for communities and project teams to interpret. They further observe that strong performance often depends on project-specific labels and feature engineering, which can hinder generalization across different contexts [7]. Their work addresses this problem through an expert-consensus methodology that combines open and axial coding with two Delphi rounds to derive a consensus-based operational definition of airdrop hunters and 15 measurable behavioral indicators, including both hunter-behavior signals and human-ness counter-evidence [7].

These observations highlight an important limitation of increasingly complex hybrid models. Although combining graph structure, behavioral information, and multimodal features can substantially improve predictive performance, the resulting latent representations and classification scores do not by themselves provide human-understandable evidence for why a wallet or group of wallets is considered suspicious. Moreover, dependence on project-specific labels may make a high-performing model difficult to transfer across campaigns or blockchain contexts. Therefore, hybrid learning provides an important performance-oriented reference point for our design, but also motivates a complementary objective: retaining the relational and behavioral information captured by these models while exposing explicit, inspectable evidence that can explain why addresses are considered coordinated Sybil actors.

2.3 Behavioral Signals for Sybil Detection

Behavioral signals identify suspicious actors by analyzing deviations in activity patterns and temporal sequences. In traditional fraud and anomaly detection, rapid, synchronized, or repet-

itive actions can provide evidence of automated or coordinated behavior [6, 12]. For example, sequential pattern mining has been used to identify fraudulent transactions by learning users' normal transaction sequences and detecting deviations from these patterns [6]. More broadly, anomaly-detection research has established behavioral deviations as an important signal for identifying abnormal activity across different application domains [12]. These observations provide methodological precedent for blockchain Sybil detection, where coordinated claim bursts, unusual transaction timing, or repetitive interaction patterns may indicate non-organic activity across wallets.

Behavioral signals are also useful in settings where reliable labels are difficult to obtain. Unlike fully supervised classifiers that require a sufficiently representative set of labeled Sybil and honest accounts, unsupervised or semi-supervised approaches can identify groups of accounts with similar or anomalous behavioral patterns without requiring exhaustive ground-truth labels. In airdrop settings, behavioral clustering can therefore serve as a candidate-generation mechanism: wallets exhibiting unusually similar activity sequences, participation timing, or transaction patterns can be grouped for further investigation or combined with graph-based evidence. This is particularly relevant to Sybil detection because the true controlling entities behind addresses are generally unobservable, making large-scale manual labeling difficult.

In blockchain airdrops, however, behavioral signals are less commonly studied as standalone Sybil detection methods and are increasingly incorporated into hybrid systems. For example, Liu et al. combine temporal and transaction-level characteristics, including address lifecycle and airdrop participation timing, with multi-layer graph representations before feature propagation, fusion, and classification [8]. This reflects a broader shift from using behavioral patterns as individual indicators toward combining them with relational and contextual evidence.

Behavioral signals alone nevertheless provide limited evidence of common ownership or coordination. Legitimate users may exhibit similar activity patterns when participating in the same campaign or interacting with the same applications, while Sybil operators can deliberately vary their timing and transaction sequences to evade simple behavioral heuristics. Behavioral analysis is therefore useful for identifying suspicious activity and generating candidate groups, but is most informative when complemented by graph-structural and transactional evidence, motivating the hybrid approaches discussed in the following section.

2.4 Resource and Metadata-Based Sybil Defenses

Resource-based defenses address Sybil attacks by making the creation or maintenance of multiple identities depend on resources that are costly, scarce, or difficult to share. Classical Sybil research identifies resource testing as a major defense family, alongside trusted certification and social-network-based approaches [10]. The foundational analysis by Douceur shows that, in the absence of a logically centralized authority, preventing Sybil attacks requires strong assumptions about the relative resources available to honest and malicious entities [2]. Depending on the system, the tested resource may include computational, storage, communication, or physical resources, as well as network-level properties or economic costs [2, 9, 11]. The central objective is therefore to make large-scale identity creation sufficiently expensive or physically constrained that an attacker cannot cheaply obtain disproportionate influence. Thus, resource-based de-

fenses primarily address the *scalability of identity creation*, rather than directly establishing that multiple identities are controlled by the same real-world entity.

Different resource-testing mechanisms impose different constraints. Newsome et al. systematically analyze Sybil attacks in sensor networks and propose several defenses, including radio-resource testing, key validation, position verification, and registration mechanisms [11]. Their radio-resource approach exploits the physical constraint that a malicious device cannot simultaneously transmit or receive on multiple channels under the assumed hardware model. More generally, resource-based defenses may exploit network-level properties such as IP addresses or network coordinates, while other approaches impose recurring economic costs on identities [10]. Margolin and Levine formalize the latter perspective by analyzing the relationship between the value of an attack and the cost of maintaining Sybil identities, and propose recurring fees as a deterrent [9]. These approaches share a common principle: Sybil resistance is obtained by constraining the resources available to an identity, rather than by reconstructing the behavioral or transactional relationships between identities.

However, resource-based evidence has two fundamental weaknesses: *evadability* and *ambiguity*. From an evasion perspective, an attacker can distribute identities across different network endpoints, devices, or access resources, weakening the assumption that a particular resource corresponds to a single underlying identity. Conversely, shared resources create the opposite problem. Multiple legitimate users may legitimately appear behind the same IP address, NAT gateway, enterprise network, university network, or other shared infrastructure. A shared resource therefore does not constitute reliable evidence of common ownership. This creates an important interpretability problem: resource-level similarity may indicate shared infrastructure, but cannot by itself explain why several identities should be attributed to the same controlling entity.

The distinction becomes particularly important in blockchain airdrop detection because classical resource-testing mechanisms operate at a different layer from the evidence available in public on-chain data. Network-level defenses observe properties of the communication environment in which identities participate, whereas public blockchain transaction records primarily expose cryptographic addresses, signed transactions, timestamps, amounts, and transaction relationships. Network attributes such as IP addresses or device-level identifiers are generally not part of the publicly observable transaction graph. Consequently, a resource-level defense may be useful as an admission-control or anti-abuse mechanism during an airdrop campaign, but it does not naturally transfer to post-hoc analysis of an on-chain wallet population.

This creates a fundamental mismatch between resource-based Sybil resistance and on-chain attribution. Even if auxiliary network metadata were available, it would provide evidence about access infrastructure rather than direct evidence of wallet ownership or coordination. Conversely, wallets controlled by the same actor can deliberately operate through different network resources and therefore appear unrelated at the resource layer. Resource signals can thus provide useful auxiliary constraints, but they do not directly capture the transactional relationships between wallets or provide sufficient evidence for attributing apparently coordinated addresses to a common controller.

In practice, resource and metadata-based mechanisms remain valuable for limiting large-

scale identity proliferation at the entry layer of an application. However, they are poorly suited to explaining coordinated wallet behavior from public blockchain data alone. This motivates the use of behavioral, graph-structural, and hybrid representations that derive evidence directly from observable transaction activity and relationships.

2.5 Comparison Table

Method Category	**Detection Mechanism**	**Interpretability**	**Performance**	**Limitations**
Resource / Metadata Based	IP addresses, device fingerprints, computational resources	Low: shared resources may come from multiple users, making attribution difficult	Medium-Low: useful as a coarse-grained first-level screening filter, but may suffer from false positives	Limited to metadata; can be evaded by distributing activity across different network resources or access points
Behavior Based	Temporal sequences, transaction bursts, repeated actions	Medium-High: suspicious activity patterns can be readily identified and explained	Medium: detects dense or anomalous activity patterns, but performance varies across different settings	Insufficient to provide strong attribution evidence independently
Graph-Structural Inference	Belief propagation over graph structure (e.g., SybilBelief, SybilFrame)	Medium: partially interpretable via graph structure, but decisions arise from propagated probabilities	Medium: effective under the structural assumptions of social-network settings, but these assumptions may not directly transfer to blockchain transaction graphs	Limited attribution interpretability; relies on probabilistic diffusion over topology rather than explicit ownership or fund-flow modeling
Hybrid Approach	Subgraph extraction + feature fusion; graph-based propagation or embedding learning followed by ML classifiers (e.g., LightGBM) or GNN-based prediction	Low–Medium: LightGBM provides partial feature-level interpretability (e.g., feature importance/SHAP), while graph propagation and GNN embeddings remain largely opaque and difficult to trace	High: captures both structural anomalies and behavioral patterns through graph context and ML-based feature fusion	Limited attribution interpretability; embedding-based reasoning is difficult to trace and provides weak explanations of cross-wallet fund flow or coordination mechanisms

2.6 Summary and Identified Research Gap

Existing Sybil detection methods have made significant progress, but each category exhibits inherent limitations in blockchain airdrop settings.

Resource- or IP-based heuristics are constrained by the absence of network metadata on-chain and can be circumvented through VPNs, proxies, or exchange-hosted wallets. They also rely on external signals rather than blockchain-native evidence and may suffer from limited attribution and false positives.

Behavior-based approaches can identify anomalous wallet activity, but they cannot fully reconstruct relationships between wallets. Similar transaction patterns alone do not necessarily indicate common control, limiting their ability to provide explicit evidence linking seemingly independent wallets to the same Sybil entity.

Graph-based structural and probabilistic inference methods, such as SybilBelief and SybilFrame, leverage connectivity patterns through mechanisms such as belief propagation or hybrid feature-structure fusion. While these approaches can capture structural relationships and propagate information across the transaction graph, graph connectivity alone does not necessarily correspond to behavioral or semantic similarity between wallets. Their outputs may therefore depend on propagated probabilistic signals whose relationship to explicit transactional behavior is difficult to trace and interpret. Hybrid approaches that further integrate graph structure with machine learning, including feature fusion and GNN-based models, can improve predictive performance but often introduce additional complexity and produce probabilistic scores or latent representations that remain difficult to interpret.

These limitations reveal a gap between predictive capability and explicit, auditable evidence of coordinated wallet control. To address this gap, this thesis proposes a hybrid framework that combines behavioral consistency with direct funding graph analysis.

The proposed framework complements resource- or IP-based heuristics by operating entirely on on-chain behavioral and funding signals, eliminating dependence on external metadata. It further complements behavior-based approaches by incorporating fund-flow structure, providing attribution evidence beyond behavioral similarity and enabling stronger identification of shared control relationships. Compared with graph-based probabilistic approaches, the framework uses observable behavioral and funding relationships directly rather than relying primarily on propagated probabilistic signals, providing a more transparent and traceable basis for identifying coordinated wallets in blockchain airdrop campaigns.

Chapter 3

Solution Concept

3.1 Modeling Rationale

The modeling of Sybil behavior in this thesis is based on an operationalization of the general Sybil concept using observable on-chain evidence. Conceptually, a Sybil attack involves a single underlying entity controlling or coordinating multiple wallets in order to act as multiple independent participants. In airdrop campaigns, such coordination is not directly observable at the entity level, but it can manifest through recurring transactional and funding patterns.

To operationalize this concept, the behavior of wallets involved in real-world airdrop campaigns is examined for recurring on-chain properties that can provide evidence of coordinated control. Two complementary patterns are particularly relevant. First, wallets involved in the same coordinated activity may exhibit similar behavioral fingerprints, including comparable funding times, transaction amounts, activity intensity, and recurring transaction sequences. Second, multiple wallet groups that appear behaviorally distinct may nevertheless share a common funding provenance, for example when funds are distributed from the same upstream source into separately executed wallet batches.

These observations motivate the operational model adopted in this thesis. Sybil behavior is therefore represented through two complementary dimensions: **behavioral coordination** and **shared funding provenance**. The first captures similarity in wallet activity, while the second provides structural evidence linking otherwise seemingly unrelated wallets to a common source of funds. Together, these dimensions provide an interpretable basis for identifying potential Sybil activity from on-chain data.

3.2 Overview of the Verification Pipeline

Figure 3.1 illustrates the operational model described above. A single underlying entity may control multiple wallet subgroups whose behavioral patterns can differ across execution batches, while shared funding provenance provides a structural link between otherwise seemingly unrelated groups.

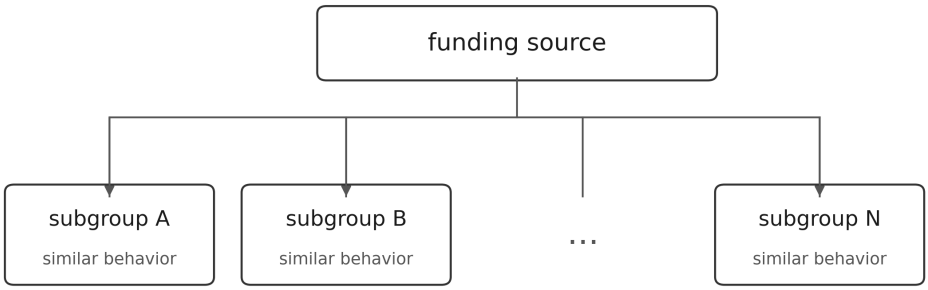


Figure 3.1: Conceptual illustration of Sybil behavior across wallet subgroups.

Based on this conceptual model, the verification process consists of six main stages, described below.

Funding-Source Backtracking and Wallet Set Expansion

Transactions are restricted to the campaign period $[t_0, t_1]$. Starting from an initial set of wallets, funding sources are recursively traced upstream to identify additional wallets that may belong to the same coordinated activity. For each wallet, the first observed funding transaction is used as the backtracking link, and all newly discovered wallets are added to the expanded suspicious set.

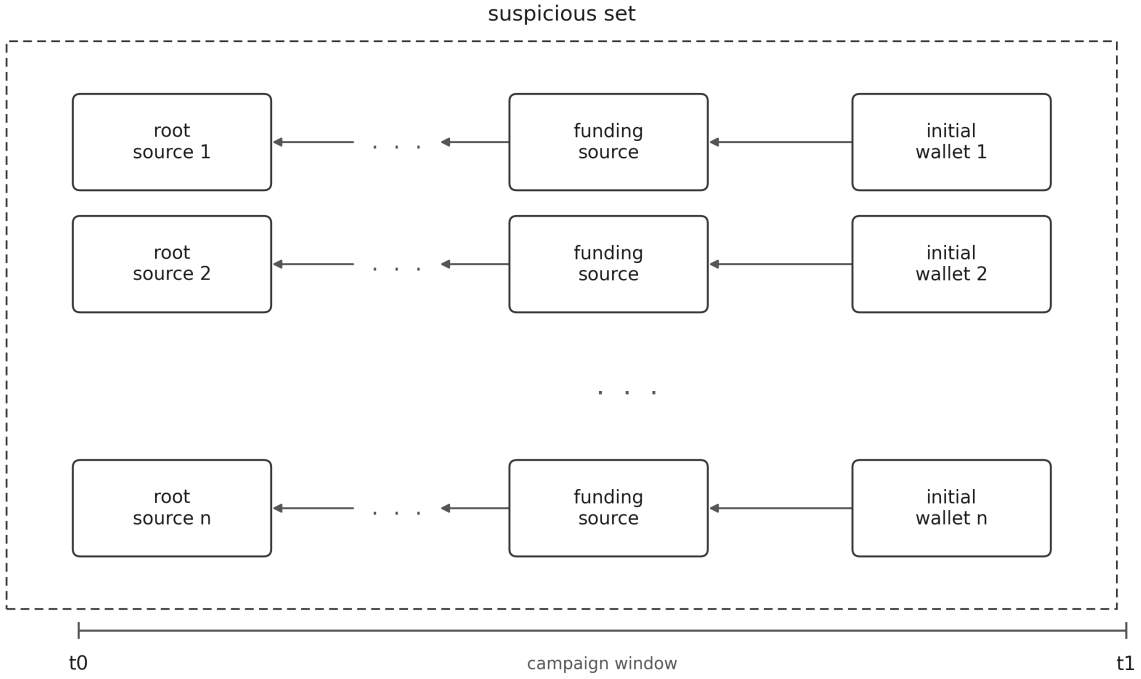


Figure 3.2: Time-bounded recursive wallet expansion through funding-source tracing.

The backtracking process is subject to several constraints to prevent irrelevant or excessive expansion. First, tracing stops when the transaction reaches the predefined start time t_0, ensuring that the analysis remains within the campaign window. Second, backtracking is terminated when a predefined maximum depth is reached, limiting the propagation of increasingly indirect funding relationships. Finally, tracing stops when a high-degree infrastructure wallet or known hub, such as a centralized exchange (CEX) or decentralized exchange (DEX), is encountered. Such wallets aggregate or redistribute funds from many potentially unrelated users, making further upstream tracing less informative for attributing common control.

These stopping conditions ensure that the expansion captures relevant funding relationships while avoiding propagation through generic infrastructure or increasingly indirect transaction paths.

Figure 3.3 shows three CEX wallets. The two outer CEX wallets are the relevant roots of their respective Sybil subtrees, while the central CEX is included for illustrative purposes only and is not reached during backtracking.

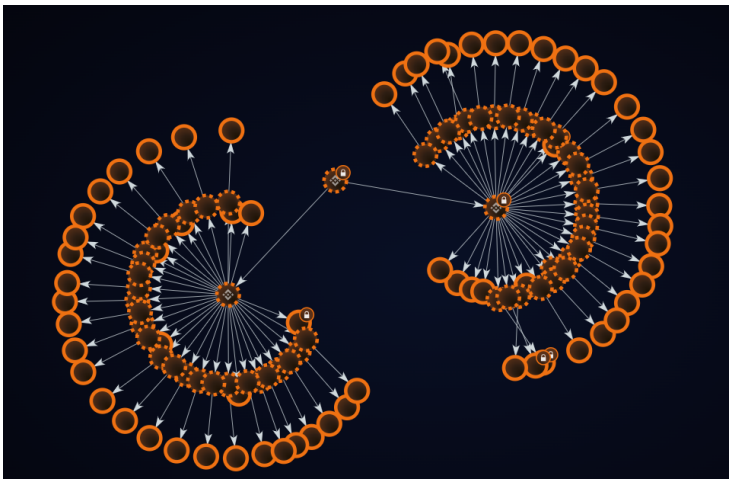


Figure 3.3: Backtracking paths terminating at CEX root wallets.

Construction of Direct Funding Graph and Funding Graph Decomposition

A directed funding graph $G = (V, E)$ is constructed from the expanded set of suspicious wallets. Here, V represents the set of wallets, and E represents directed edges corresponding to funding transactions, where an edge (u, v) indicates that wallet u is the first funder of wallet v. Depth-First Search (DFS) identifies independent connected components, denoted as ICC_k, each representing wallets linked through funding relationships. Components define the potential scope of a Sybil structure and reduce the complexity of global analysis.

Behavioral Feature Extraction and Process Variants

Each wallet $w_i \in \text{ICC}_k$ is represented by:

- **Behavioral features:** total transaction count, first funding time and first funding amount.
- **Process variant:** an ordered sequences of relevant actions (first funding transfer, approval, swap, claim) capturing automation patterns.

Behavioral features and process variants are jointly used to compute wallet pairwise similarities, forming the similarity matrix that serves as the basis for clustering.

Weighted Decay Mechanism (See Section 3.3)

The Weighted Decay Mechanism is used to compute pairwise similarity between any two wallets, incorporating both behavioral features and process variant sequences. The resulting pairwise similarities are used to construct a similarity matrix, which is further transformed into a distance matrix for clustering.

Behavioral Clustering via DBSCAN

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) is a density-based clustering algorithm that groups data points according to local density without requiring the number

of clusters to be specified in advance. It is particularly suitable for settings where clusters may have irregular structures and where noise or weakly related points are expected to exist.

In the context of blockchain Sybil detection, DBSCAN is well-suited because coordinated wallet groups can form dense regions in the behavioral similarity space, while unrelated or weakly related wallets may be classified as noise. This allows the method to separate tightly coordinated Sybil entities from independent or weakly related wallets without requiring prior knowledge of the number of clusters.

Let ICC_k denote the k-th independent connected component, and let $w_i, w_j \in \mathrm{ICC}_k$ denote wallets within that component. Let $Sim(w_i, w_j)$ denote the pairwise similarity between wallets w_i and w_j, computed using the Weighted Decay Mechanism. For each wallet w_i, its ϵ-neighborhood is defined as

$$N_\epsilon(w_i) = \{w_j \in \mathrm{ICC}_k \mid Sim(w_i, w_j) \geq \epsilon\}.$$

A wallet w_i is considered a core point if its ϵ-neighborhood contains at least minPts wallets. DBSCAN then forms clusters through density-connected core points and their reachable neighboring wallets. The resulting l-th cluster within ICC_k is denoted by

$$\mathrm{Cluster}_{k,l} \subseteq \mathrm{ICC}_k,$$

where k indexes the independent connected component and l indexes the cluster within that component.

Intuitively, each $\mathrm{Cluster}_{k,l}$ represents a group of wallets exhibiting sufficiently dense behavioral similarity and therefore constitutes a potential Sybil subgroup. Wallets that do not satisfy the density requirements of any cluster are treated as noise and are considered in subsequent noise interpretation steps.

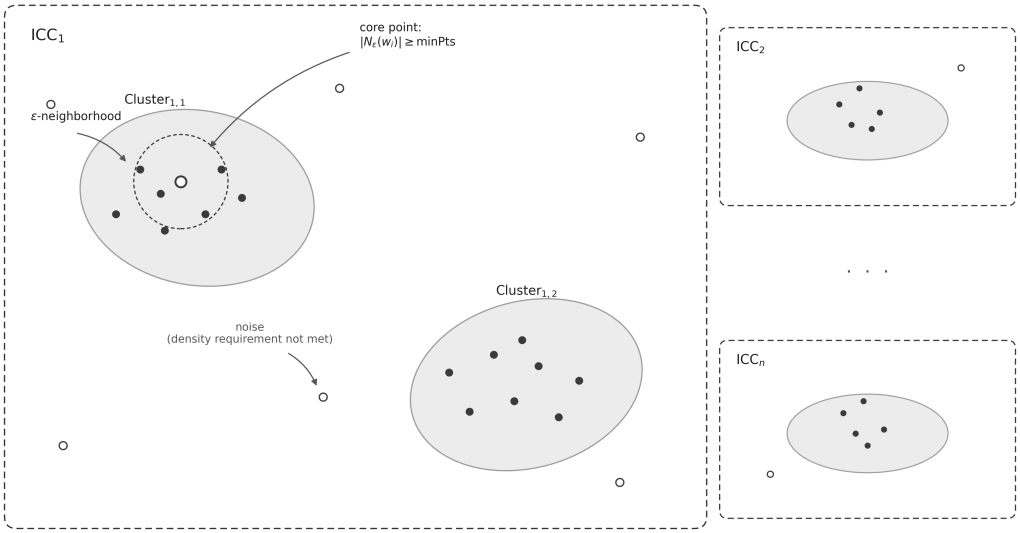


Figure 3.4: DBSCAN clustering within an independent connected component (ICC_1): dense regions form Sybil-subgroup candidates ($\mathrm{Cluster}_{k,l}$), while sparse wallets are labeled as noise. Additional components ($\mathrm{ICC}_2, \ldots, \mathrm{ICC}_n$) are processed identically.

3.2.1 Extension: Noise Interpretation via contamination inference

Density-based clustering methods such as DBSCAN are widely used to detect Sybil groups due to their ability to identify arbitrarily shaped clusters in transaction graphs. However, a fundamental limitation of DBSCAN is that it produces a set of unassigned points, referred to as *noise nodes*.

In the context of Sybil detection, these noise wallets are not necessarily benign. They may correspond to (i) borderline Sybil accounts with irregular behavior, (ii) structurally weak Sybil nodes with sparse connectivity, or (iii) nodes excluded due to local density variations.

Therefore, instead of discarding these noise nodes, we aim to further characterize and interpret them by measuring their contamination exposure to known Sybil structures.

We model this as a bounded contamination inference problem over the transaction graph, integrating structural and behavioral evidence as complementary signals.

Graph-based contamination inference

We measure structural contamination by modeling the influence of known Sybil wallets within the local graph neighborhood of a target wallet. For each noise wallet w, the funding graph is explored bidirectionally: upstream, the traversal follows the unique funding lineage of w, while downstream, it explores all reachable successor branches subject to the defined stopping conditions and maximum hop distance. Let $N_k(w)$ denote the set of known Sybil wallets encountered at a shortest graph distance of k hops from w during this traversal. The Sybil exposure intensity is then computed by aggregating these Sybil wallets across hop-distance shells, with the contribution of each shell decreasing quadratically with distance:

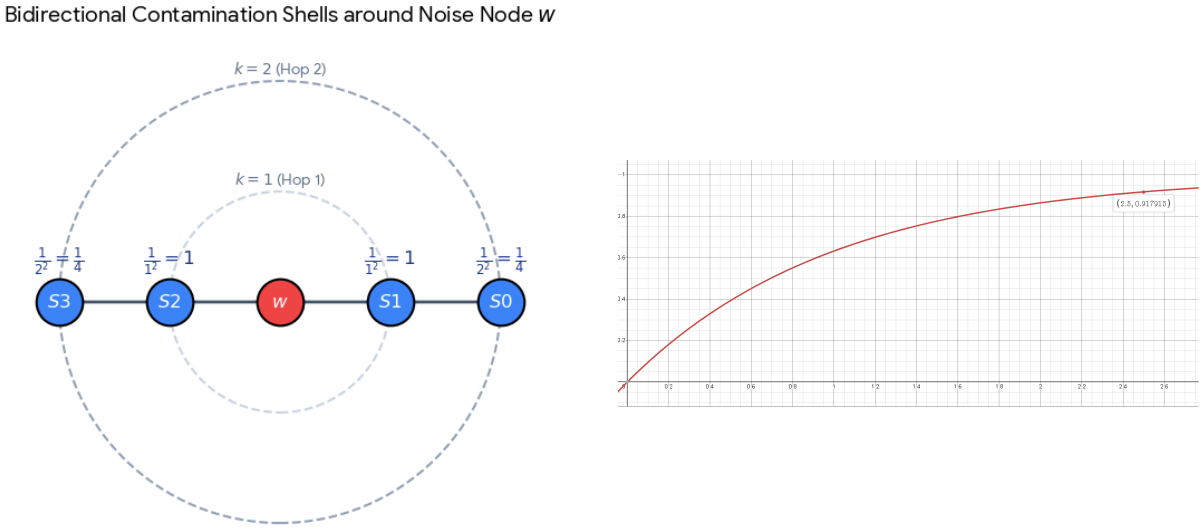
$$x_{\text{graph}}(w) = \sum_{k=1}^{K} \frac{|N_k(w)|}{k^2} \tag{3.1}$$

where K denotes the maximum considered hop distance. The resulting exposure intensity is transformed into a bounded contamination score:

$$E_{\text{graph}}(w) = 1 - e^{-x_{\text{graph}}(w)}. \tag{3.2}$$

Intuition. The graph-based contamination score can be viewed as a radial aggregation of Sybil influence around the target wallet. Sybil wallets located one hop away contribute most strongly, while those in successive hop-distance shells contribute progressively less due to the quadratic decay term $1/k^2$. At each distance, the contributions of all Sybil wallets within the same shell are aggregated, allowing the score to account for both proximity to Sybil activity and the number of Sybil wallets surrounding the target.

The exponential transformation maps the unbounded exposure intensity to the range $[0, 1)$, providing a normalized measure of structural contamination while reducing the effect of very large exposure values in densely connected regions.



In the left figure, we compute contamination score by aggregating neighbor influence across distance shells. Nodes directly connected to known Sybil wallets contribute the strongest signal, while indirect connections are progressively discounted. This produces a scalar structural exposure value $x_{\text{graph}}(w)$, which summarizes both direct and multi-hop proximity to Sybil behavior. In the illustrated case, the aggregation yields $x_{\text{graph}}(w) = 2.5$, indicating substantial structural proximity within the fund flow graph.

We then map this unbounded exposure into a bounded contamination score using an exponential transformation $E_{\text{graph}}(w) = 1 - e^{-x}$. The intuition behind this choice is that it preserves strong sensitivity at low exposure while preventing unbounded growth for highly connected nodes. When $x \leq 1$, corresponding to either a single strong connection or multiple weak Sybil-linked connections, the score already reaches approximately 0.63. This reflects a meaningful but not decisive signal — it indicates elevated suspicion, where the node may be related to Sybil activity but still requires further verification. As x increases, the score rises rapidly at first, capturing compounding evidence, and then gradually saturates toward 1, ensuring diminishing returns from excessive connectivity and avoiding domination by hub effects.

In the case shown in the left figure where $x_{\text{graph}}(w) = 2.5$, the resulting score is $E_{\text{graph}}(w) \approx 0.918$. This indicates a strong structural correlation with known Sybil wallets in the funding graph, suggesting a high likelihood of contamination in the local neighborhood structure.

Behavioral contamination

We aggregate behavioral similarity between the noise wallet and known Sybil wallets.

Behavioral exposure is defined as the cumulative similarity to all Sybil wallets whose similarity (calculated with weighted decay mechanism) exceeds a threshold τ:

$$x_{\text{beh}}(w) = \sum_{\substack{u \in W_{\text{sybil}} \\ sim(w,u) \geq \tau}} sim(w, u) \tag{3.3}$$

$$E_{\text{beh}}(w) = 1 - e^{-x_{\text{beh}}(w)} \tag{3.4}$$

Intuition. Behavioral contamination captures semantic alignment between a target wallet and known Sybil behavioral patterns. Instead of relying on structural connectivity, it measures how strongly a wallet resembles Sybil entities in terms of transaction behavior. The threshold τ filters weak or noisy similarities, ensuring only meaningful behavioral matches contribute to the signal. The cumulative form reflects the idea that repeated alignment with multiple Sybil behaviors increases suspicion, while the exponential mapping introduces diminishing returns to prevent redundant or highly correlated behavioral patterns from dominating the score.

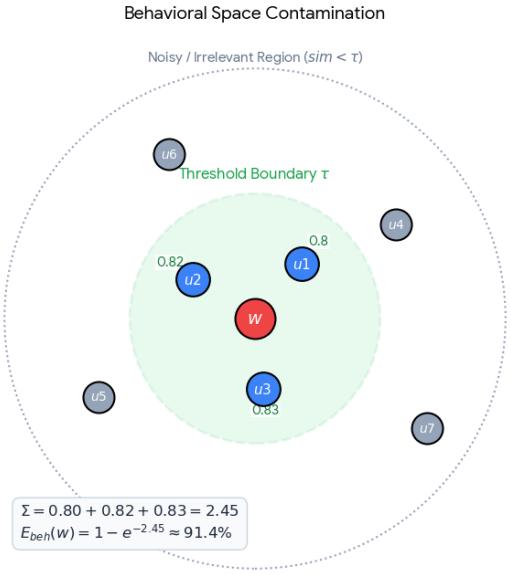


Figure 3.5: The Figure illustrates the behavioral contamination framework for a target noise node w. The threshold boundary represents τ, which segments the feature space into two regions. Wallets in the outer noisy region (u_4, u_5, u_6, u_7) are excluded because their similarity falls below the filter criteria $(sim(w, u) < \tau)$. Conversely, the nodes inside the perimeter (u_1, u_2, u_3) satisfy $sim(w, u) \geq \tau$. Their individual similarity metrics (0.80, 0.82, and 0.83) are additively aggregated to compute the total behavioral intensity $x_{\text{beh}}(w) = 2.45$. Finally, this unbounded sum is passed through an exponential mapping function to yield the normalized exposure score $E_{\text{beh}}(w) = 1 - e^{-2.45} \approx 91.4\%$.

Unified evidence fusion

We model Sybil detection as the aggregation of heterogeneous evidence sources into a unified risk representation. Each view produces a non-negative evidence intensity, capturing the strength of anomalous signals from different observational domains.

$$x(w) = x_{\text{graph}}(w) + x_{\text{beh}}(w) \tag{3.5}$$

where $x_{\text{graph}}(w)$ measures structural propagation strength in the funding graph, and $x_{\text{beh}}(w)$ captures behavioral similarity to known Sybil-like entities.

We then map the aggregated intensity into a bounded risk score:

$$S(w) = 1 - e^{-x(w)} \tag{3.6}$$

Interpretation. The function $S(w)$ is not intended to represent a calibrated probability. Instead, it defines a normalized unified risk score that preserves the monotonic accumulation of evidence across heterogeneous sources. The exponential transformation ensures that (i) the score is bounded in $[0, 1)$, (ii) additional evidence always increases risk, and (iii) multiple weak signals can collectively produce strong risk when aggregated.

This formulation provides a principled alternative to heuristic weighting schemes by ensuring consistent scale and additive interpretability across different evidence modalities, while avoiding explicit probabilistic calibration assumptions.

Multi-view Contamination Score and Conservative Aggregation

In addition to the unified log-odds score, we retain the individual contamination views:

$$C_{\text{graph}}(w) = E_{\text{graph}}(w), \quad C_{\text{beh}}(w) = E_{\text{beh}}(w), \quad C_{\text{uni}}(w) = S(w) \tag{3.7}$$

Because different evidence channels may provide stronger signals under different structural conditions, we adopt a conservative aggregation rule:

$$C(w) = \max\left(C_{\text{graph}}(w), C_{\text{beh}}(w), C_{\text{uni}}(w)\right) \tag{3.8}$$

Intuition. The multi-view design preserves three distinct perspectives on contamination: structural proximity, behavioral similarity, and their joint evidence. Rather than averaging these signals, the maximum operator assigns the final score according to the strongest available evidence. This conservative rule avoids diluting a strong signal from one view when other views provide weaker evidence, which is particularly useful when different Sybil structures produce asymmetric signals across the behavioral and graph views.

3.2.2 Summary

The solution concept provides a complete pipeline for interpretable detection of coordinated Sybil wallets:

1. Expand suspicious wallet set based on funding history.
2. Decompose the funding graph into independent connected components to reduce the search space for subsequent similarity computation and clustering.
3. Represent wallets with behavioral vectors and process variants.
4. Compute pairwise similarity using the Weighted Decay Mechanism (see Section 3.3).
5. Cluster wallets using DBSCAN.
6. Noise Interpretation via contamination inference framework

3.3 Weighted Decay Mechanism

Traditional similarity measures can be categorized into two main regimes: magnitude-sensitive metrics (e.g., Euclidean distance) and magnitude-invariant metrics (e.g., cosine similarity). Both fail in this setting, but for different reasons.

Euclidean distance is highly sensitive to feature scale, where dominant features can overwhelm weaker behavioral signals even after normalization. Cosine similarity, on the other hand, removes magnitude information through L2 normalization and focuses only on angular relationships. As a result, short-term coordinated behaviors in dense Sybil clusters may become compressed, indistinguishable, or misclustered.

To address these limitations, we propose the *Weighted Decay Mechanism*, which computes pairwise wallet similarity using feature-specific weighting and decay-based similarity functions. This produces a similarity matrix for downstream clustering while preserving temporal synchronization and interpretable feature-level contributions.

3.3.1 Overall Similarity Definition

For any two wallets i and j, we define the overall similarity as a weighted combination of behavioral vector similarity and process variant similarity:

$$\mathrm{Sim}(i,j) = w_{\mathrm{behv}} \cdot \mathrm{Sim}_{\mathrm{behv}}(i,j) + w_{\mathrm{pv}} \cdot \mathrm{Sim}_{\mathrm{pv}}(i,j), \tag{3.9}$$

where:

- $\mathrm{Sim}_{\mathrm{behv}}(i,j)$ is the weighted decay similarity between the behavioral vectors of wallets i and j,
- $\mathrm{Sim}_{\mathrm{pv}}(i,j)$ is the similarity between process variant sequences, calculated using the Levenshtein distance to account for order and sequence differences,
- w_{behv} and w_{pv} are the relative weights for the behavioral vector and process variant sequence, respectively.

3.3.2 Behavioral Vector Similarity

The behavioral vector similarity is defined as:

$$\mathrm{Sim}_{\mathrm{behv}}(i,j) = \sum_{f=1}^{N} w_f \cdot \mathrm{decay}_f\Big(|x_i^{(f)} - x_j^{(f)}|\Big) \tag{3.10}$$

where:

- N is the number of behavioral features,
- w_f is the weight assigned to feature f, reflecting its importance for detecting coordinated behavior,
- $\mathrm{decay}_f(\cdot)$ is a feature-specific decay function that reduces the contribution of large differences, emphasizing meaningful short-term coordination.

3.3.3 Process Variant Sequence Similarity

The process variant similarity, $\text{Sim}_{\text{pv}}(i, j)$, uses the Levenshtein distance:

$$\text{Sim}_{\text{pv}}(i, j) = 1 - \frac{\text{Lev}(s_i, s_j)}{\max(|s_i|, |s_j|)}, \tag{3.11}$$

where s_i and s_j are the process variant sequences of wallets i and j, respectively. This ensures the similarity captures both order and differences in the sequence of airdrop-related actions.

3.3.4 Comparison of Different Similarity Measures

Example 1: Limitations of Euclidean Distance-Based Similarity

We evaluate both similarity formulations on a minimal two-wallet setting. Since the methods operate on pairwise relations, this setup directly exposes edge cases and sensitivity to feature scaling and temporal modeling. The Euclidean method is purely magnitude-based in a unified feature space, where dominance of a single feature can skew the representation and suppress contributions from other weaker features.

Wallet	Tx	Timestamp	Fund
w1	10	29083901	0.01
w2	11	29083989	0.01

Euclidean Distance-Based Similarity Construction We define feature vectors as:

$$\mathbf{x}_i = (tx_i, fund_i, time_i)$$

To address scale heterogeneity, Min-Max normalization is applied:

$$\tilde{\mathbf{x}}_i = \text{MinMax}(\mathbf{x}_i)$$

Pairwise Euclidean distance is computed:

$$d(i, j) = \|\tilde{\mathbf{x}}_i - \tilde{\mathbf{x}}_j\|_2$$

The distance matrix is normalized to $[0, 1]$:

$$\hat{d}(i, j) = \frac{d(i, j) - d_{\min}}{d_{\max} - d_{\min}}$$

Similarity is then obtained via inversion:

$$S_{\text{euc}}(i, j) = 1 - \hat{d}(i, j)$$

Weighted Decay Similarity The proposed weighted decay mechanism uses a weighted combination of feature-level similarities, including exponential decay over raw temporal differences.

(A) Euclidean Distance-Based Similarity

	w1	w2
w1	1.000	0.000
w2	0.000	1.000

(B)Weighted Decay Similarity

	w1	w2
w1	1.000	0.636
w2	0.636	1.000

Figure 3.6: Comparison between Euclidean Distance-Based Similarity and the proposed weighted decay mechanism on the same wallet pair.

Result Interpretation.
The Euclidean formulation produces a near-binary outcome, assigning zero off-diagonal similarity despite highly aligned behavioral attributes (total tx count and first fund amount).

This is because Euclidean distance is purely magnitude-based in a unified geometric space, where all feature dimensions are collapsed into a single L2 norm. As a result, it does not preserve feature-wise behavioral semantics, but instead measures only global numeric displacement.

In contrast, the weighted decay formulation preserves partial similarity (0.636), reflecting consistent transaction-level and funding-level alignment while accounting for temporal proximity through exponential decay. This yields a smoother and more stable similarity structure under small behavioral perturbations, avoiding the binary collapse effect observed in Euclidean distance-based formulations.

Example 2: Limitations of Cosine Similarity

We construct a minimal pair of wallets sharing the same temporal position but exhibiting different levels of transactional activity and funding behavior.

Wallet	Tx	Timestamp	Fund
w1	10	29083969	0.1
w2	1	29083969	0.01

Each wallet is represented in a feature space defined as:

$$x = (\log(1 + tx),\ time_{norm},\ \log(1 + fund))$$

where tx denotes transaction count, $fund$ denotes funding amount, and $time_{norm}$ is the min-max normalized block height defined as:

$$time_{norm} = \frac{timestamp - \min(timestamp)}{\max(timestamp) - \min(timestamp)}.$$

After feature construction, L2 normalization is applied, and cosine similarity is computed between wallet representations.
The resulting cosine similarity matrix is:

	$w1$	$w2$
$w1$	1	0.99967814
$w2$	0.99967814	1

In contrast, under the proposed weighted decay mechanism, the same pair of wallets yields a more discriminative similarity structure:

	$w1$	$w2$
$w1$	1.000	0.550
$w2$	0.550	1.000

Under this formulation, the same pair of wallets is no longer treated as nearly identical, reflecting the influence of both behavioral and temporal discrepancies captured by the decay-based similarity function.
Despite a tenfold difference in both transaction count and funding amount, cosine similarity produces values extremely close to 1, indicating near-collinear vectors in the normalized feature space.
This occurs because cosine similarity measures angular relationships between vectors while discarding absolute magnitude information. Consequently, wallets with substantially different behavioral intensities may still exhibit high similarity when their feature ratios remain consistent. In contrast, the weighted decay mechanism incorporates both feature-level differences and temporal decay, resulting in a more sensitive similarity measure that preserves meaningful separation under heterogeneous behavioral scales.
This example demonstrates that cosine similarity is not sensitive even to substantial variations in behavioral scale, whereas the proposed weighted decay formulation maintains discriminative power under the same conditions.

Example 3: Temporal Compression Issue with Normalization

The table shows three groups of transactions exhibiting clear temporal separation in block numbers. Batch 1 and Batch 2 belong to the same underlying project and are temporally adjacent within a relatively short window (on the order of tens of minutes to approximately half an hour). However, they correspond to different execution rounds within the same campaign rather than a single synchronized burst.
In contrast, Batch 3 is both temporally distant and associated with a different project, representing a separate behavioral process.
When Min-Max normalization is applied over the full temporal range, the relative gap between Batch 1 and Batch 2 becomes compressed. As a result, Euclidean distance becomes less sensitive to fine-grained temporal differences, reducing its ability to distinguish closely spaced execution rounds.

Address	Tx	Timestamp	Fund	Batch
0x079dd1...0897	4	29362247	0.001188	1
0xc36d2a...4edc	4	29362243	0.001269	1
0x046f05...2c66	4	29362236	0.001158	1
0x687c59...9543	4	29362236	0.001104	1
0x36b092...071d	4	29362235	0.001126	1
0x0f6618...02d4	4	29362210	0.001186	2
0x303d6d...f0a2	4	29362205	0.001151	2
0xeddefc...af8a	4	29362215	0.001236	2
0x2655b2...9038	4	29362206	0.001228	2
0xcef2e5...d682	4	29362210	0.001125	2
0xedf95d...3ff0	2	29083971	0.01	3
0x292c7d...f8ba	2	29083972	0.01	3
0x24d9f7...c225	2	29083972	0.01	3
0xe7c09d...34ce	2	29083970	0.01	3
0xb46052...6d5e	2	29083973	0.01	3

This highlights a key limitation of global normalization: when the overall time span is large, local temporal variations are effectively flattened. In Sybil detection settings, however, short-term coordination patterns often carry important signals of structured or automated behavior.
Figure 3.7 illustrates this difference. Under Euclidean distance with Min-Max normalization, Batch 1 and Batch 2 are partially merged despite being distinct execution rounds. With the proposed weighted decay mechanism, the two batches remain clearly separable while preserving intra-group coherence.

3.3.5 Discussion

The results highlight fundamental limitations of standard similarity measures in Sybil detection with coordinated short-term behaviors. Euclidean distance assumes a globally consistent geometric scale, leading to sensitivity to normalization and suppression of localized temporal structure. Cosine similarity assumes directional sufficiency, discarding magnitude information and thus reducing sensitivity to behavioral intensity differences. These limitations reflect fundamental constraints of global embedding-based similarity models rather than implementation-specific design choices.

In contrast, the weighted decay mechanism computes feature-level similarities before aggregation and incorporates temporal decay, enabling joint modeling of behavioral intensity and temporal coordination. Each feature contributes independently to the final similarity through explicit weighting, resulting in an interpretable decomposition of similarity structure. This mitigates both scale collapse and magnitude loss while preserving discriminative structure under heterogeneous behavioral patterns.

Overall, the proposed method enables more reliable clustering of coordinated behavioral patterns in Sybil detection settings.

```
=== DBSCAN Labels (Euclidean + MinMax) ===
Wallet 0x079dd142f2ae648a952de968a15cff38d420b897 => Cluster 0
Wallet 0xc36d2a36ad4d01b11e5aa72bb969d0470aff4edc => Cluster 0
Wallet 0x046f05cb76e75d4d6a39ac3b978be02c30cf2c66 => Cluster 0
Wallet 0x687c591b11992180c79b26840db3f0918539543d => Cluster 0
Wallet 0x36b09293beca924cc4515b4ca0623a20e470c71d => Cluster 0
Wallet 0x0f661802c8ec7f3be9f76109e242d8efcbb502d4 => Cluster 0
Wallet 0x303d6d6d1029247bd4cdee66059b64392ef3f0a2 => Cluster 0
Wallet 0xeddefc061ba45f9500f66e5d0be6d7149e10af8a => Cluster 0
Wallet 0x2655b22493e5c4df83c4e16a383be2c29469038d => Cluster 0
Wallet 0xcef2e56a5bc14b2a6a1ff2a316b50ef6ab5d682e => Cluster 0
Wallet 0xedf95d57559d42a40d60235c4b705b0b93383ff0 => Cluster 1
Wallet 0x292c7d4b8cd2c4f8a8fd79d54653676c7374f8ba => Cluster 1
Wallet 0x24d9f7dd917d675a3ef1f95a33a820a71fc2254d => Cluster 1
Wallet 0xe7c09d21ccd5903ea2935f2d46e5141b4aa34ce1 => Cluster 1
Wallet 0xb460525c88004246aed64a4702999c8e2a566d5e => Cluster 1
```

```
=== DBSCAN Labels ===
Wallet 0x079dd142f2ae648a952de968a15cff38d420b897 => Cluster 0
Wallet 0xc36d2a36ad4d01b11e5aa72bb969d0470aff4edc => Cluster 0
Wallet 0x046f05cb76e75d4d6a39ac3b978be02c30cf2c66 => Cluster 0
Wallet 0x687c591b11992180c79b26840db3f0918539543d => Cluster 0
Wallet 0x36b09293beca924cc4515b4ca0623a20e470c71d => Cluster 0
Wallet 0x0f661802c8ec7f3be9f76109e242d8efcbb502d4 => Cluster 1
Wallet 0x303d6d6d1029247bd4cdee66059b64392ef3f0a2 => Cluster 1
Wallet 0xeddefc061ba45f9500f66e5d0be6d7149e10af8a => Cluster 1
Wallet 0x2655b22493e5c4df83c4e16a383be2c29469038d => Cluster 1
Wallet 0xcef2e56a5bc14b2a6a1ff2a316b50ef6ab5d682e => Cluster 1
Wallet 0xedf95d57559d42a40d60235c4b705b0b93383ff0 => Cluster 2
Wallet 0x292c7d4b8cd2c4f8a8fd79d54653676c7374f8ba => Cluster 2
Wallet 0x24d9f7dd917d675a3ef1f95a33a820a71fc2254d => Cluster 2
Wallet 0xe7c09d21ccd5903ea2935f2d46e5141b4aa34ce1 => Cluster 2
Wallet 0xb460525c88004246aed64a4702999c8e2a566d5e => Cluster 2
```

Figure 3.7: Comparison between Euclidean distance with Min-Max normalization (left) and the proposed weighted decay mechanism (right).

3.4 Chapter Summary

This chapter operationalizes the proposed model of Sybil behavior through the combination of behavioral coordination and shared funding provenance. Wallet activity is represented through behavioral features and process variants, while funding relationships are captured in a directed transaction graph. Together, these representations allow coordinated activity to be examined through both behavioral similarity and shared fund-flow structure.

The resulting pipeline combines wallet-level behavioral representation, weighted pairwise similarity, and density-based clustering within independent funding components. The Weighted Decay Mechanism captures both transaction-level intensity and process-level coordination, while DBSCAN identifies dense behavioral subgroups without requiring the number of groups to be specified in advance. The funding graph provides an additional structural perspective for interpreting how these subgroups are related through their observed funding provenance.

The chapter also extends the analysis to wallets that remain unresolved by the primary clustering stage. Rather than treating DBSCAN noise as inherently irrelevant, the contamination inference framework re-evaluates these wallets using their local structural exposure and behavioral similarity to known Sybil activity. This produces complementary contamination scores that preserve the underlying evidentiary dimensions and provide an interpretable basis for further investigation.

Overall, the chapter establishes a combined behavioral and funding-based representation for identifying coordinated wallet activity, while extending the analysis to sparse or borderline cases that fall outside the primary behavioral clusters.

The main methodological contributions introduced in this chapter are:

- A **Weighted Decay Mechanism** for measuring pairwise wallet similarity through feature-wise aggregation, temporal decay, and process-level sequence similarity.
- A **contamination inference framework** for interpreting DBSCAN noise through complementary structural and behavioral evidence, allowing otherwise unresolved wallets to be evaluated for potential association with known Sybil activity.

Chapter 4

Implementation

This chapter presents the implementation details of the proposed verification framework, including dataset construction, wallet labeling strategy, event filtering, wallet set expansion, feature extraction, similarity computation, clustering, and system pipeline design.

4.1 Dataset

We construct a real-world dataset consisting of **428 unique wallets** collected from three airdrop campaigns across different blockchain ecosystems:

- **Project 1 (MYX)** – BNB Chain
- **Project 2 (Apriori)** – BNB Chain
- **Project 3 (IRYS)** – Ethereum

All wallets are extracted from **real on-chain transaction data**, ensuring that the dataset reflects authentic blockchain activity rather than synthetic or simulated environments. The dataset is therefore suitable for studying adversarial coordination behaviors in realistic Sybil attack scenarios.

4.1.1 Dataset Reliability and Construction Rationale

Unlike benchmark datasets with definitive ground-truth labels, real-world Sybil detection lacks a universally observable ground truth because wallet ownership and off-chain coordination are not directly visible on-chain. Rather than relying on a single labeling criterion, this thesis constructs a high-confidence reference dataset through the convergence of multiple sources of evidence that are external to the proposed detection pipeline.

Sybil labels are established through three complementary sources:

- **On-chain forensic evidence:** manual examination of transaction behavior and funding relationships, including synchronized activity, shared funding sources, recurring execution patterns, and coherent fund-flow structures;

- **Independent third-party intelligence:** corroboration through publicly available wallet labels, investigations, and cluster analyses from professional blockchain intelligence and forensic platforms, including Nansen and Bubblemaps;
- **External investigative evidence:** publicly documented investigations, expert analyses, and practitioner findings, including relevant reports and discussions published on X (Twitter) and other public sources.

A wallet is treated as a high-confidence Sybil reference only when the available evidence provides sufficient convergence across these sources, rather than solely because it satisfies the behavioral or graph criteria of the proposed framework. This separation is important for evaluation because the reference labels are established independently of the Weighted Decay Mechanism, DBSCAN clustering, and contamination inference procedure, thereby reducing the risk of circular validation.

The resulting labels are therefore treated as *high-confidence reference labels* rather than absolute ground truth. This approach provides a stronger basis for evaluating the proposed framework while acknowledging that residual uncertainty cannot be completely eliminated in an adversarial and only partially observable blockchain environment.

4.1.2 Case Study Nature of the Dataset

We emphasize that this dataset is not intended to serve as a fully representative benchmark of all blockchain activity. Instead, it is designed as a **targeted case study dataset**, focusing on known or highly suspected Sybil clusters.

This design choice allows us to:

- isolate realistic coordination patterns in adversarial environments,
- evaluate detection mechanisms under high-signal, high-risk scenarios,
- study behavioral and structural contamination signals in controlled but real-world conditions.

As such, the dataset prioritizes *label confidence and behavioral richness* over large-scale coverage.

4.2 Data Cleaning and Event Filtering

Raw blockchain transaction histories include many irrelevant actions. Each wallet's transactions are converted into a filtered event log retaining only events relevant to airdrop participation:

- First funding transaction: timestamp, amount, sender
- Approval transactions of the specified tokens
- Claim or batch claim transactions of the specified tokens
- Swap transactions of the specified tokens

- Transfer or batch transfer transactions of the specified tokens within the campaign period

Specified tokens may include the native gas token, airdropped tokens, or other relevant tokens for behavioral analysis. All transactions are stored as per-wallet CSV files to facilitate manual backtracking while preserving logical equivalence to automated API-based collection.

4.3 Wallet Set Expansion and Funding Graph Construction

The initial wallet set is expanded by recursively tracing upstream funding relationships. For each wallet, the first observed funding transaction within the campaign period is identified, and its funding source is added to the wallet set. The process continues recursively until a predefined stopping condition is reached, such as the campaign start time t_0 or a known hub wallet such as a centralized exchange (CEX).

The resulting relationships are represented as a directed funding graph $G = (V, E)$, where the node set V contains the initial wallets and all subsequently discovered funding sources, and each directed edge $(f, w) \in E$ represents a direct funding relationship from funder f to wallet w.

Algorithm 1 Wallet Set Expansion and Funding Graph Construction

Require: Initial wallet set W_{init}, campaign start date t_0, hub wallet set H
Ensure: Expanded wallet set V and directed funding graph $G = (V, E)$

```
Initialize V ← W_init, E ← ∅
for all wallet w ∈ W_init do
    Identify the first funding source f of w
    while f exists and f ∉ H and date(f → w) ≥ t_0 do
        Add f to V
        Add edge (f, w) to E
        w ← f                                   ▷ Move one hop upstream
        Identify the first funding source f of w
    end while
end for
return V, G = (V, E)
```

4.4 Feature Extraction

Each wallet is represented using behavioral and structural features:

- Total number of transactions (as defined in Section 4.3)
- First funding timestamp (minutes, UTC)
- First funding amount
- Ordered sequence of relevant actions (process variants)

For each wallet, these features form a behavioral vector used for similarity computation and clustering.

4.4.1 Pairwise Similarity Function

Behavioral similarity is computed using a weighted combination of feature-wise similarity functions. For two wallets w_i and w_j with behavioral vectors $\mathbf{x}_i = (tx_i, t_i, amt_i)$ and $\mathbf{x}_j = (tx_j, t_j, amt_j)$, we define:

Feature-wise Similarity Functions

- **Transaction count similarity:**

$$s_{\text{tx}}(i,j) = \min\left(1, \frac{\min(tx_i, tx_j)}{\max(tx_i, tx_j)}\right) \tag{4.1}$$

- **First funding amount similarity:**

$$s_{\text{amt}}(i,j) = \begin{cases} 1, & \text{if } \max(amt_i, amt_j) \leq \delta \\ \min\left(1, \frac{\min(amt_i, amt_j)}{\max(amt_i, amt_j)}\right), & \text{otherwise} \end{cases} \tag{4.2}$$

  where δ is a small threshold (e.g., 0.002) to treat negligible amounts as identical.

- **First funding time similarity:**

$$s_{\text{time}}(i,j) = \begin{cases} 1, & \text{if } |t_i - t_j| \leq \delta \\ \exp\left(-\frac{|t_i - t_j| - \delta}{\lambda}\right), & \text{otherwise} \end{cases} \tag{4.3}$$

  where δ defines the temporal tolerance window within which two wallets are considered highly similar in time (in our experiments $\delta = 5$ minutes), and λ controls the decay rate of similarity beyond this window.

Behavioral Similarity The overall behavioral similarity is defined as:

$$S_{\text{beh}}(i,j) = w_1 \cdot s_{\text{tx}}(i,j) + w_2 \cdot s_{\text{amt}}(i,j) + w_3 \cdot s_{\text{time}}(i,j) \tag{4.4}$$

where $w_1 + w_2 + w_3 = 1$ (default: equal weights).

Variant Similarity Variant similarity is computed using normalized Levenshtein ratio:

$$S_{\text{var}}(i,j) = \text{Lev}(v_i, v_j) \tag{4.5}$$

where v_i, v_j are the process variant sequences.

Final Similarity The overall wallet similarity is:

$$S(i,j) = \alpha \cdot S_{\text{var}}(i,j) + (1-\alpha) \cdot S_{\text{beh}}(i,j) \tag{4.6}$$

where $\alpha \in [0,1]$ controls the importance of variant similarity, in our case, it is set as 0.5 ,meaning variant similarity is as important as behavior vector similarity.

Parameter Interpretation The decay parameter λ in the time similarity function controls temporal sensitivity. In this work, $\lambda = 2.2$, which ensures that:

- Wallets funded within 5 minutes are treated as identical ($s_{\text{time}} = 1$)
- Similarity decays smoothly afterward
- After approximately 15 minutes, similarity approaches 0

This reflects the intuition that Sybil wallets are typically funded within short time intervals, while larger gaps indicate weaker coordination.

4.5 Clustering

- **DBSCAN:** applied to the distance matrix $D = 1 - S$, where S is the pairwise similarity matrix.
- Default parameters: $\epsilon = 0.1$, minPts $= 3$.
- Intuition: each dense cluster corresponds to a highly coordinated subgroup, potentially a Sybil entity.

4.6 Bidirectional Contamination Inference Implementation

To operationalize the proposed contamination model, we implement a bidirectional graph traversal procedure that computes structural and behavioral contamination scores for each DBSCAN noise wallet. For each target noise wallet w, the funding graph is explored relative to w as the center of a local neighborhood. The upstream direction follows the wallet's unique funding lineage, while the downstream direction may branch into multiple successor wallets. The traversal is bounded by predefined stopping conditions and a maximum hop depth. Known Sybil wallets encountered during the traversal contribute to the structural contamination score according to their hop distance from w.

4.6.1 Implementation Overview

For each noise wallet $w \in W_{\text{noise}}$, the procedure consists of four main operations:

- **Backward traversal (funding lineage):** Starting from w, follow its unique upstream funder recursively until a terminal node, a predefined hub such as a centralized exchange (CEX), or the maximum hop depth $H_{\max}$ is reached.
- **Forward traversal (fund distribution):** Starting from w, recursively explore all downstream successor wallets that it funds. Because a wallet may fund multiple successors, the forward traversal may branch into multiple paths. Traversal continues until leaf nodes, stopping nodes, or the maximum hop depth $H_{\max}$ is reached.

- **Sybil encounter scoring:** During both traversals, known Sybil wallets are recorded according to their hop distance k from w. The number of encountered Sybil wallets within each distance shell contributes to the graph exposure intensity using the quadratic decay factor $1/k^2$.
- **Behavioral aggregation:** Independently of the graph traversal, the target wallet is compared against known Sybil wallets using the Weighted Decay similarity measure. Only similarities satisfying the threshold $\tau = 0.8$ contribute to the behavioral exposure score.

The resulting structural and behavioral evidence is transformed into bounded contamination scores according to the formulations defined in Section 3.2.1.

Algorithm 2 Backward Funding-Lineage Traversal

Require: Target noise wallet w, known Sybil set W_{sybil}, funding graph $G = (V, E)$, maximum hop depth $H_{\max}$

Ensure: Hop-distance Sybil sets $N_k(w)$

Initialize $N_k(w) \leftarrow \emptyset$ for $k = 1, \ldots, H_{\max}$
Initialize visited set $\mathcal{V} \leftarrow \{w\}$
$u \leftarrow w$
for $k \leftarrow 1$ **to** $H_{\max}$ **do**
    Identify the unique upstream funder p of u
    if p does not exist **or** p is a stopping node **then**
        break
    end if
    if $p \in \mathcal{V}$ **then**
        break
    end if
    Add p to $\mathcal{V}$
    if $p \in W_{\text{sybil}}$ **then**
        Add p to $N_k(w)$
    end if
    $u \leftarrow p$
end for
return $\{N_1(w), \ldots, N_{H_{\max}}(w)\}$

Algorithm 3 Forward Fund-Distribution Traversal

Require: Target noise wallet w, known Sybil set W_{sybil}, funding graph $G = (V, E)$, maximum hop depth $H_{\max}$

Ensure: Hop-distance Sybil sets $N_k(w)$

Initialize queue $Q \leftarrow \{(w, 0)\}$
Initialize visited set $\mathcal{V} \leftarrow \{w\}$
Initialize $N_k(w) \leftarrow \emptyset$ for $k = 1, \ldots, H_{\max}$
while Q is not empty **do**
    Remove (u, d) from Q
    if $d = H_{\max}$ **then**
        continue
    end if
    for all successors v of u in G **do**
        if $v \notin \mathcal{V}$ **then**
            $d' \leftarrow d + 1$
            Add v to $\mathcal{V}$
            if $v \in W_{\text{sybil}}$ **then**
                Add v to $N_{d'}(w)$
            end if
            if v is not a stopping node **then**
                Add (v, d') to Q
            end if
        end if
    end for
end while
return $\{N_1(w), \ldots, N_{H_{\max}}(w)\}$

4.6.2 Contamination Score Computation

The detailed mathematical formulation of the graph-based contamination score, behavioral contamination score, and unified evidence score is provided in Section 3.2.1. The implementation applies these formulations independently to each DBSCAN noise wallet after its local graph neighborhood has been explored.

For the graph view, the encountered Sybil wallets are grouped into hop-distance shells $N_k(w)$ relative to the target wallet. The shell counts are aggregated using the quadratic decay factor $1/k^2$ and transformed into the bounded score $E_{\text{graph}}(w)$. For the behavioral view, similarities between the target wallet and known Sybil wallets are accumulated only when they satisfy the threshold $\tau = 0.8$, after which the resulting intensity is mapped to $E_{\text{beh}}(w)$.

Algorithm 4 Behavioral Contamination and Multi-View Score Aggregation

Require: Target noise wallet w, known Sybil set W_{sybil}, structural score $E_{\text{graph}}(w)$, similarity threshold τ

Ensure: Final contamination score $C(w)$

Behavioral contamination
Initialize $x_{\text{beh}}(w) \leftarrow 0$
for all $u \in W_{\text{sybil}}$ **do**
    Compute $Sim(w, u)$ using the Weighted Decay Mechanism
    if $Sim(w, u) \geq \tau$ **then**
        $x_{\text{beh}}(w) \leftarrow x_{\text{beh}}(w) + Sim(w, u)$
    end if
end for
$E_{\text{beh}}(w) \leftarrow 1 - e^{-x_{\text{beh}}(w)}$

Unified evidence
$S(w) \leftarrow 1 - e^{-(x_{\text{graph}}(w) + x_{\text{beh}}(w))}$

Multi-view aggregation
$C_{\text{graph}}(w) \leftarrow E_{\text{graph}}(w)$
$C_{\text{beh}}(w) \leftarrow E_{\text{beh}}(w)$
$C_{\text{uni}}(w) \leftarrow S(w)$
$C(w) \leftarrow \max\left(C_{\text{graph}}(w), C_{\text{beh}}(w), C_{\text{uni}}(w)\right)$
return $C(w)$

4.6.3 Final Score Definition

For each noise wallet, the structural, behavioral, and unified contamination scores are retained as

$$C_{\text{graph}}(w) = E_{\text{graph}}(w), \quad C_{\text{beh}}(w) = E_{\text{beh}}(w), \quad C_{\text{uni}}(w) = S(w). \tag{4.7}$$

The final contamination score is defined using a conservative maximum aggregation:

$$C(w) = \max\left(C_{\text{graph}}(w), C_{\text{beh}}(w), C_{\text{uni}}(w)\right). \tag{4.8}$$

The maximum operator retains the strongest evidence available across the three views, avoiding dilution of a strong signal by weaker complementary views.

4.7 System Implementation

The proposed framework is implemented using Python and JavaScript-based components. The main tools used for data processing, graph analysis, and clustering are summarized below:

- **Pandas:** data cleaning, transformation, and transaction event-log management.
- **D3.js:** funding graph representation, visualization, and graph traversal operations.
- **Scikit-learn:** DBSCAN clustering and related clustering utilities.

Frontend is implemented using Next.js to visualize clusters, funding graphs, and potential sybil entities.

4.8 Chapter Summary

This chapter translates the solution concept developed in Chapter 3 into a complete Sybil verification pipeline and its corresponding implementation. The proposed behavioral and funding-based model is operationalized on real blockchain data through dataset construction, feature extraction, similarity computation, graph-based analysis, density-based clustering, and contamination inference. Together, these components turn the conceptual model into an executable verification pipeline that can be evaluated systematically on real-world wallet activity.

Chapter 5

Experiments and Results

5.0.1 Definition of Verified Sybil Wallets

In this work, a *verified Sybil wallet* is defined as a wallet that satisfies both behavioral coordination and funding provenance consistency. Specifically, it must belong to a behaviorally coherent subgroup identified by clustering and be traceable to a shared top-level funding source via direct funding graph backtracking.

DBSCAN noise wallets are not considered verified due to insufficient local density evidence. However, the proposed noise interpretation extension can still provide insights into how strongly these nodes correlate with existing Sybil wallets.

In total, the experiments involve 428 wallets across three projects, including 21 control wallets. Each subgroup represents a highly synchronized batch within a Sybil cluster.

5.1 Project A: MYX (BSC)

Project MYX contains 151 wallets forming a star-like sybil cluster. The wallets are split into 5 subgroups.

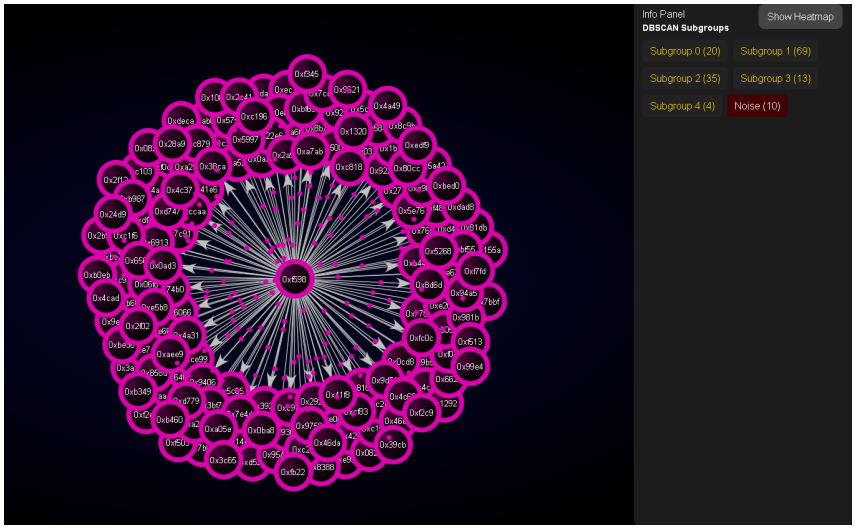


Figure 5.1: Representative sybil cluster in Project MYX.

Cluster Summary

- Total wallets: 151
- Number of subgroups: 5
- Topology: star-like
- Number of noise wallets: 10
- Number of wallets excluding noise: 151 - 10 = 141

5.2 Project B: Apriori (BSC)

Project Apriori contains two tree-like sybil clusters.

Tree 1

- Total wallets: 55
- Number of subgroups: 11
- Topology: tree-like
- Number of noise wallets: 8
- Number of wallets excluding noise: 55 - 8 = 47

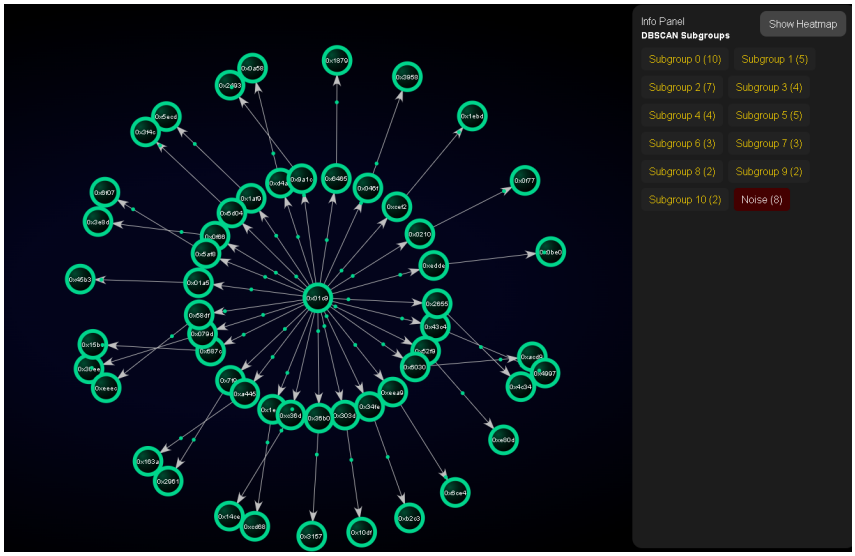


Figure 5.2: Tree 1 sybil cluster in Project Apriori.

Tree 2

- Total wallets: 49
- Number of subgroups: 9
- Topology: tree-like
- Number of noise wallets: 9
- Number of wallets excluding noise: 49 - 9 = 40

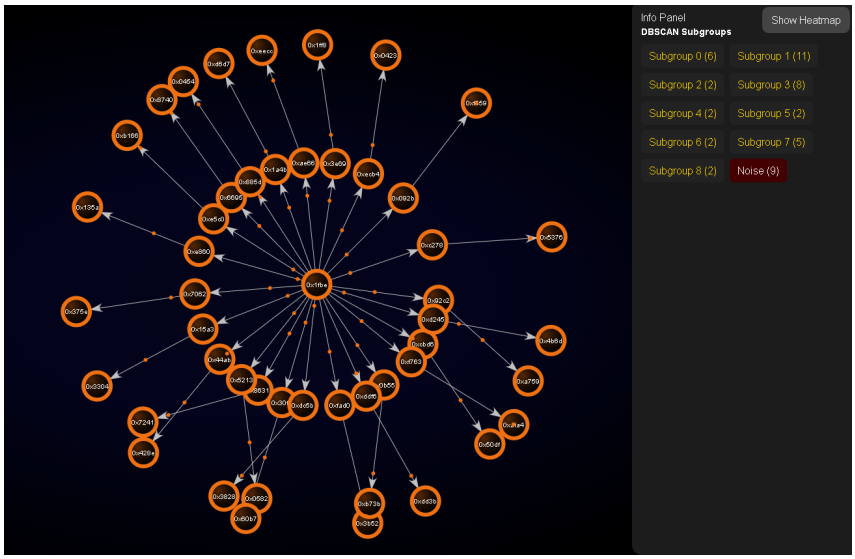


Figure 5.3: Tree 2 sybil cluster in Project Apriori.

5.3 Project C: IRYS (ETH)

Project IRYS contains 152 wallets forming a representative sybil cluster with 6 subgroups.

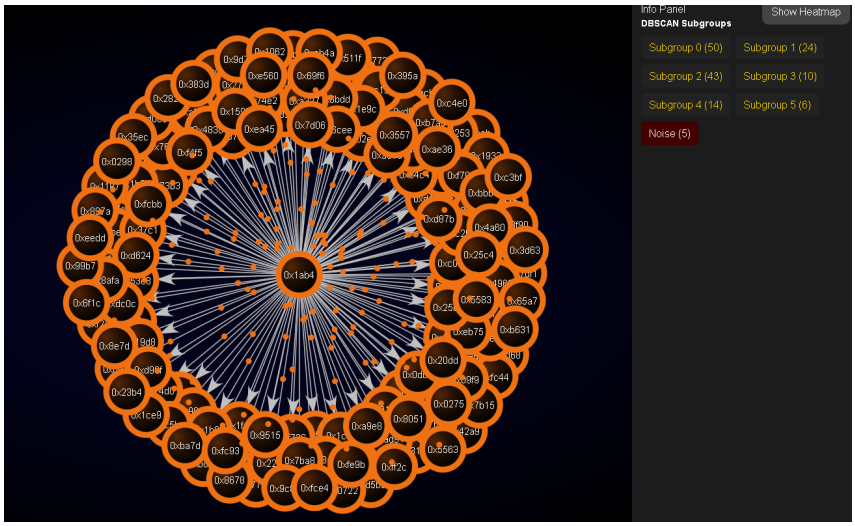


Figure 5.4: Representative sybil cluster in Project IRYS.

Cluster Summary

- Total wallets: 152
- Number of subgroups: 6
- Topology: star-like
- Number of noise wallets: 5
- Number of wallets excluding noise: 152 - 5 = 147

5.3.1 Experimental Results: Contamination Inference

Configuration. Following the cohesion analysis in Table 6.8, we adopt eps = 0.1 as the primary DBSCAN configuration for downstream analysis, yielding a total of 32 noise wallets across the three case-study campaigns considered below. All 32 wallets originate from airdrop campaigns in which the surrounding Sybil cluster(s) had been independently identified; each noise wallet was excluded from its corresponding sub-cluster solely due to falling short of the behavioral similarity threshold used by DBSCAN, and therefore carries a ground-truth Sybil label independent of the clustering outcome.

Categorization scheme. For each noise wallet w, we report the raw exposure intensities $x_{\text{graph}}(w)$, $x_{\text{beh}}(w)$, the corresponding bounded contamination scores $C_{\text{graph}}(w)$, $C_{\text{beh}}(w)$, $C_{\text{uni}}(w)$, and the final ensemble score $C(w)$. Based on which of $x_{\text{graph}}(w)$ and $x_{\text{beh}}(w)$ are non-zero, we assign each wallet to one of four categories:

- **Graph-dominant**: $x_{\text{graph}}(w) > 0$, $x_{\text{beh}}(w) = 0$
- **Behavior-dominant**: $x_{\text{graph}}(w) = 0$, $x_{\text{beh}}(w) > 0$
- **Unified-signal**: $x_{\text{graph}}(w) > 0$ and $x_{\text{beh}}(w) > 0$
- **insufficient evidence**: $x_{\text{graph}}(w) = x_{\text{beh}}(w) = 0$

Table 5.1: Category distribution of noise wallets across the three case studies (eps=0.1)

Case	Total Noise	Graph-dom.	Behavior-dom.	Unified	insufficient
Case 1: IRYS (ETH)	5	1	1	0	3
Case 2: MYX (BSC)	10	1	6	0	3
Case 3: Apriori (BSC), Cluster 1	9	1	2	6	0
Case 3: Apriori (BSC), Cluster 2	8	1	2	5	0
Total	**32**	**4**	**11**	**11**	**6**

Of the 32 noise wallets, 26 receive a non-zero final risk score ($C(w) > 0$), while 6 wallets receive $C(w) = 0$ despite carrying a ground-truth Sybil label.

Case 1: IRYS (ETH)

Table 5.2 reports representative wallets from each observed category for the single Sybil cluster in the IRYS campaign (5 noise wallets total; full listing in Appendix Table 1).

Table 5.2: Case 1 (IRYS): representative noise wallets by category

Wallet	x_{graph}	x_{beh}	C_{graph}	C_{beh}	C_{uni}	$C(w)$
Graph-dominant						
0x1ab4...8f23	147.0000	0.0000	1.0000	0.0000	1.0000	1.0000
Behavior-dominant						
0x0722...a9c6	0.0000	121.8004	0.0000	1.0000	1.0000	1.0000
insufficient-evidence						
0x0db7...d2ca	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

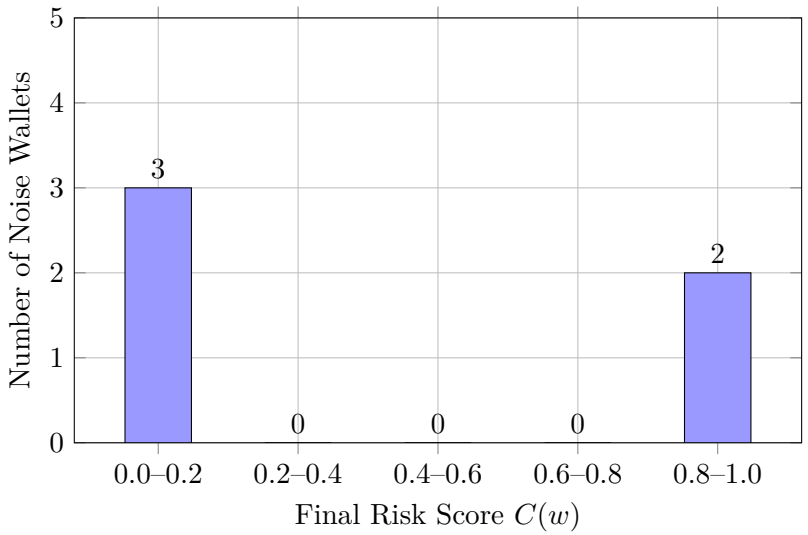


Figure 5.5: Distribution of final risk score $C(w)$ for Case 1 (IRYS, ETH), $n = 5$ noise wallets.

Case 2: MYX (BSC)

Table 5.3 reports representative wallets for the single Sybil cluster in the MYX campaign (10 noise wallets total; full listing in Appendix Table 2).

Table 5.3: Case 2 (MYX): representative noise wallets by category

Wallet	x_{graph}	x_{beh}	C_{graph}	C_{beh}	C_{uni}	$C(w)$
Graph-dominant						
0xf598...b115	141.0000	0.0000	1.0000	0.0000	1.0000	1.0000
Behavior-dominant (range across 6 wallets)						
0xbf65...b315	0.0000	43.6952	0.0000	1.0000	1.0000	1.0000
0x8388...493d	0.0000	2.4052	0.0000	0.9098	0.9098	0.9098
insufficient evidence						
0x1b70...2620	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

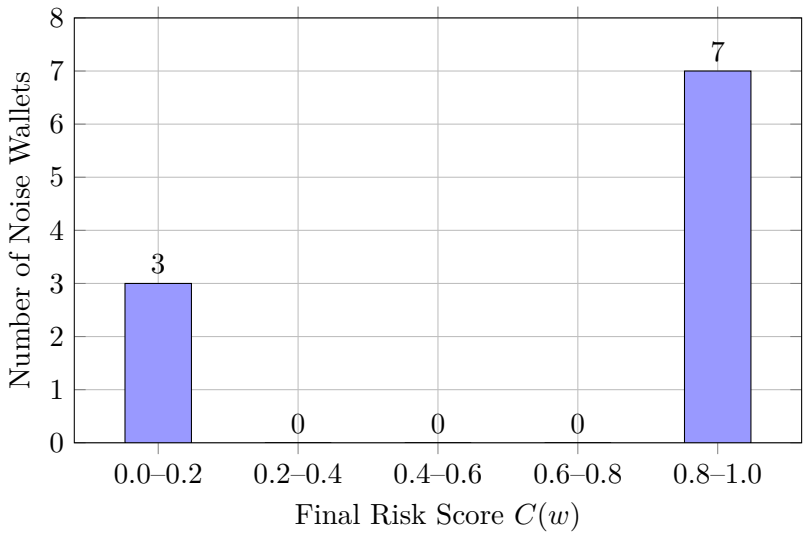


Figure 5.6: Distribution of final risk score $C(w)$ for Case 2 (MYX, BSC), $n = 10$ noise wallets.

Case 3: Apriori (BSC)

The Apriori campaign contains two independently identified Sybil clusters. Table 5.4 reports representative wallets from both clusters (17 noise wallets total; full listing in Appendix Tables 3–4).

Table 5.4: Case 3 (Apriori): representative noise wallets by category

Wallet	x_{graph}	x_{beh}	C_{graph}	C_{beh}	C_{uni}	$C(w)$
Cluster 1 – Graph-dominant						
0x1fbe...508b	22.7500	0.0000	1.0000	0.0000	1.0000	1.0000
Cluster 1 – Behavior-dominant						
0x0423...9958	0.0000	12.5000	0.0000	1.0000	1.0000	1.0000
Cluster 1 – Combined-signal						
0x1a4b...067d	1.0000	9.1667	0.6321	0.9999	1.0000	1.0000
0xdc5b...ae1b	1.0000	5.0448	0.6321	0.9936	0.9976	0.9976
Cluster 2 – Graph-dominant						
0x01c9...0bb0	28.2500	0.0000	1.0000	0.0000	1.0000	1.0000
Cluster 2 – Behavior-dominant						
0xacd9...3db6	0.0000	3.3610	0.0000	0.9653	0.9653	0.9653
Cluster 2 – Combined-signal						
0x2961...b1c2	1.0000	3.5133	0.6321	0.9702	0.9890	0.9890

For this wallet, the graph contamination score is derived from backward-path (incoming) influence (0.6321), whereas all other combined-signal wallets in this case derive their graph score from forward-path (outgoing) influence.

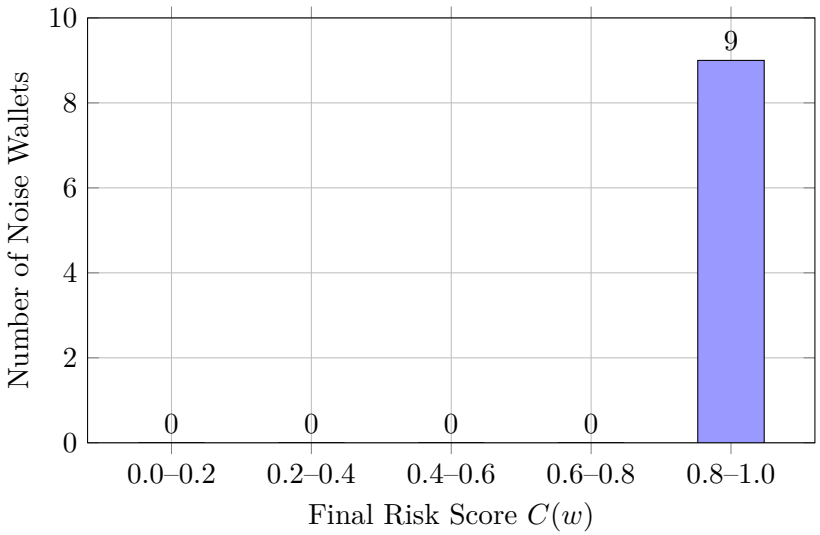


Figure 5.7: Distribution of final risk score $C(w)$ for Case 3 (Apriori, BSC), Cluster 1, $n = 9$ noise wallets.

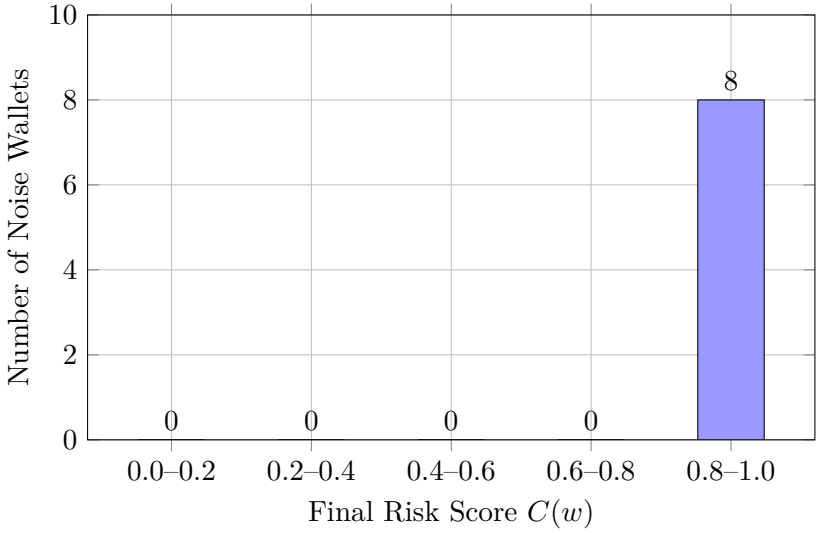


Figure 5.8: Distribution of final risk score $C(w)$ for Case 3 (Apriori, BSC), Cluster 2, $n = 8$ noise wallets.

5.4 Control Wallets

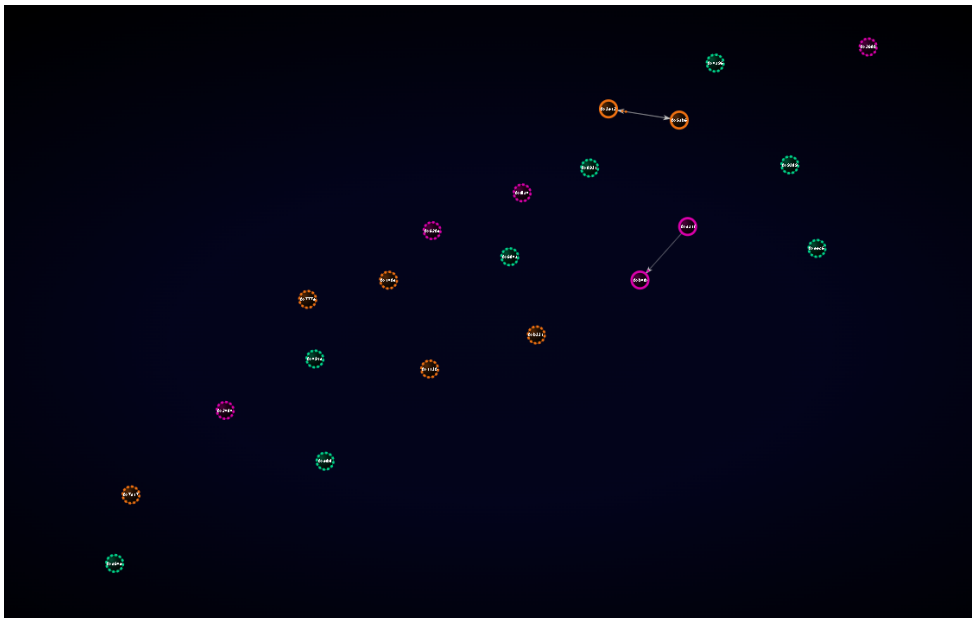


Figure 5.9: The 21 control wallets remain isolated from verified Sybil clusters, forming individual wallets or small clusters of two wallets.

A total of 21 control wallets were included in the experiments to assess whether the proposed pipeline incorrectly groups benign wallets into Sybil clusters. The control wallets were processed through the same funding-graph construction, behavioral feature extraction, similarity calculation, and DBSCAN clustering procedures applied to the Sybil candidates. As shown in Figure 5.9, the resulting control-wallet structures remain sparse: most wallets appear as individual points, while a small number form clusters of only two wallets. No control wallet is connected through an observed funding path to any verified Sybil cluster.

The clustering results therefore show no formation of large, densely coordinated groups among the control wallets. In total, none of the 21 control wallets was identified as part of a verified Sybil cluster, resulting in an observed false-positive rate of $0/21 = 0\%$. This result indicates that the proposed pipeline did not generate false positives on the evaluated control set. However, it should not be interpreted as evidence that the framework can universally achieve a 0% false-positive rate across other datasets or blockchain ecosystems. The result is specific to the evaluated control wallets and experimental setting, as discussed further in the evaluation chapter.

5.5 Chapter Summary

This chapter provides the empirical basis for assessing the proposed Sybil verification pipeline on real-world blockchain campaigns. The experiments examine the resulting Sybil structures and behavioral clusters, the separation of verified control wallets, and the ability of contamination inference to provide additional evidence for wallets left unresolved by the primary clustering stage. These results provide empirical support for the behavioral and funding-based modeling developed in the previous chapters, while also revealing the conditions and cases that motivate the limitations and further analysis discussed in the following chapter.

Chapter 6

Evaluation

6.1 Evaluation Goals

The evaluation aims to:

1. Validate the performance of the proposed pipeline in identifying potential Sybil wallets.
2. Measure the effectiveness of the noise interpretation extension.
3. Assess false positives on control wallets.
4. Measure the clustering quality of the detected groups.
5. Test the sensitivity and robustness of the pipeline under different parameter settings.
6. Provide supervised baseline comparisons using XGBoost and LightGBM to establish a predictive reference and contextualize the proposed framework's interpretable, evidence-based approach.

6.2 Verification Rate

Verification Rate: The proportion of high-confidence labelled Sybil wallets that are successfully verified by the proposed pipeline. A wallet is considered verified if it is not classified as DBSCAN noise and satisfies both the behavioural similarity and funding-relationship criteria defined by the framework.

$$\text{Verification Rate} = \frac{N_{\text{verified}}}{N_{\text{Sybil labels}}}$$

where N_{verified} denotes the number of high-confidence Sybil-labelled wallets that satisfy all verification criteria, while $N_{\text{Sybil labels}}$ denotes the total number of high-confidence Sybil labels. Wallets classified as DBSCAN noise are therefore not considered verified.

Comments:

- MYX (BSC): 141/151 = 93.4% Sybil verification rate.
- Apriori Tree 1 (BSC): 47/55 = 85.5% Sybil verification rate.

Project	Total	Noise	Verified Sybil
MYX (BSC)	151	10	141
Apriori Tree 1 (BSC)	55	8	47
Apriori Tree 2 (BSC)	49	9	40
IRYS (ETH)	152	5	147
Total	407	32	375

Table 6.1: Summary of verified Sybil wallets across projects. Noise wallets are excluded from the verified Sybil set.

- Apriori Tree 2 (BSC): 40/49 = 81.6% Sybil verification rate.
- IRYS (ETH): 147/152 = 96.7% Sybil verification rate.
- Overall: 375/407 = 92.1% Sybil verification rate across all datasets.

Project	Total Wallets	Verified Sybil	Verification Rate
MYX (BSC)	151	141	93.4%
Apriori Tree 1 (BSC)	55	47	85.5%
Apriori Tree 2 (BSC)	49	40	81.6%
IRYS (ETH)	152	147	96.7%
Overall	407	375	92.1%

Table 6.2: Verification rate of Sybil detection across datasets. Noise is excluded from evaluation.

Comments:

- Verification performance is consistent across datasets, ranging from 81.6% to 96.7%.
- Overall verification rate is 92.1%, indicating strong detection performance under the proposed pipeline under current dataset.

6.3 Cluster Quality: Cohesion and Separation

Since the proposed framework relies on behavioral similarity to form Sybil subgroups, cluster quality is evaluated to verify whether the resulting groups are behaviorally coherent while remaining sufficiently distinguishable from one another. Cluster quality is evaluated using subgroup cohesion and separation. We first measure intra-subgroup cohesion. Cohesion is defined as the mean pairwise similarity within each subgroup:

$$\mathrm{Sim}_{pv}^{subg} = \frac{2}{N(N-1)} \sum_{i<j} \mathrm{Sim}_{pv}(i,j),$$

where N is the number of wallets in the subgroup.

6.3.1 Cohesion

Observations:

- Cohesion remains high across the majority of subgroups, indicating that the wallets grouped together exhibit strong behavioral similarity.

Table 6.3: Cluster and Subgroup Cohesion Summary for $\epsilon = 0.1$

Cluster	**Subgroup**	**Cohesion**
0	0	0.9383
0	1	0.9358
0	2	0.9076
0	3	0.9160
0	4	1.0000
1	0	0.9030
1	1	1.0000
1	2	1.0000
1	3	1.0000
1	4	1.0000
1	5	1.0000
1	6	1.0000
1	7	1.0000
1	8	1.0000
2	0	0.9967
2	1	0.9243
2	2	0.9963
2	3	0.9962
2	4	0.9460
2	5	0.9965
3	0	0.8908
3	1	1.0000
3	2	1.0000
3	3	1.0000
3	4	0.9206
3	5	1.0000
3	6	0.9711
3	7	1.0000
3	8	0.9797
3	9	1.0000
3	10	1.0000
Mean		**0.9748**

- Several subgroups achieve perfect cohesion of 1.0000, suggesting that the corresponding wallets exhibit identical or highly consistent behavioral patterns.
- The lowest cohesion is observed for Cluster 3, Subgroup 0, with a value of 0.8908, while the remaining subgroups generally maintain cohesion above 0.90.
- The mean cohesion across all 31 subgroups is **0.9748**, demonstrating a high overall level of intra-subgroup behavioral consistency.

6.3.2 Subgroup Separation Analysis

Pairwise distances between subgroups within each cluster are computed across multiple behavioral and structural dimensions. Since the dimensions represent different types of information, each dimension is treated according to its underlying data type.

For continuous vector dimensions, such as Time, Total Transactions (TX), and Fund, the average distance of subgroup i to all other subgroups is defined as:

$$d_i^{\text{vector}} = \frac{1}{N-1} \sum_{\substack{j=0 \\ j \neq i}}^{N-1} d_{i,j}^{\text{vector}}, \tag{6.1}$$

where N is the total number of subgroups and $d_{i,j}^{\text{vector}}$ denotes the pairwise distance between subgroups i and j for the corresponding dimension. For dimensions such as Fund, Time, and TX, the distance is computed using standardized differences or an appropriate vector distance metric.

For the process Variant dimension, which consists of discrete strings or textual information, the Levenshtein similarity ratio is used to measure the similarity between two subgroups. The average Variant similarity for subgroup i is given by:

$$s_i^{\text{variant}} = \frac{1}{N-1} \sum_{\substack{j=0 \\ j \neq i}}^{N-1} \text{Lev}(v_i, v_j), \tag{6.2}$$

where v_i and v_j denote the Variant strings of subgroups i and j, respectively. The Levenshtein similarity ratio $\text{Lev}(\cdot, \cdot)$ ranges from 0 to 1, with higher values indicating greater structural similarity.

Composite Similarity. To capture overall behavioral and structural similarity, a composite similarity is defined at the pairwise level by combining process Variant similarity with behavioral similarity. For each pair of subgroups (i, j):

$$s_{i,j}^{\text{comp}} = \lambda \cdot \text{Lev}(v_i, v_j) + (1 - \lambda) \cdot s_{i,j}^{\text{behavior}}, \tag{6.3}$$

where $\lambda \in [0, 1]$ controls the relative contribution of process Variant similarity. In this work, $\lambda = 0.5$, giving equal importance to structural and behavioral similarity.

The behavioral similarity is computed over the three continuous dimensions, namely TX, Fund, and Time:

$$s_{i,j}^{\text{behavior}} = \sum_k w_k s_{i,j}^{(k)}, \qquad \sum_k w_k = 1, \tag{6.4}$$

where

$$k \in \{\text{TX}, \text{Fund}, \text{Time}\},$$

and equal weights are used:

$$w_k = \frac{1}{3}.$$

The subgroup-level composite distance is then obtained by averaging the pairwise composite similarities and converting the resulting similarity into a distance:

$$d_i^{\text{comp}} = 1 - \frac{1}{N-1} \sum_{\substack{j=0 \\ j \neq i}}^{N-1} s_{i,j}^{\text{comp}}. \tag{6.5}$$

Thus, higher composite values indicate greater separation from the other subgroups in the same cluster, whereas lower values indicate greater overall similarity. This formulation combines structural similarity captured by process Variants with behavioral similarity captured by Time, TX, and Fund.

Cluster 0

Table 6.4: Cluster 0: Average subgroup distances across behavioral and Variant dimensions.

Subgroup	Time	TX	Fund	Variant	Composite
0	0.274	0.779	0.060	0.780	0.575
1	1.000	0.827	0.240	0.867	0.778
2	0.382	0.742	0.060	0.680	0.537
3	0.294	0.739	0.060	0.676	0.520
4	0.335	0.849	0.060	0.794	0.604

Cluster 0 exhibits substantial variation across TX and Variant distances. Subgroup 1 has the highest composite distance (0.778), reflecting its relatively large separation in Time, TX, Fund, and Variant dimensions. Subgroups 3 and 2 exhibit lower composite distances of 0.520 and 0.537, respectively, indicating greater similarity to the remaining subgroups. Overall, both behavioral activity and process Variant differences contribute to the separation observed within this cluster.

Cluster 1

Table 6.5: Cluster 1: Average subgroup distances across behavioral and Variant dimensions.

Subgroup	Time	TX	Fund	Variant	Composite
0	1.000	0.000	0.021	0.000	0.170
1	1.000	0.000	0.021	0.000	0.170
2	0.991	0.000	0.019	0.000	0.168
3	1.000	0.000	0.021	0.000	0.170
4	1.000	0.000	0.020	0.000	0.170
5	0.991	0.000	0.021	0.000	0.169

Cluster 1 has uniformly low composite distances, ranging from 0.168 to 0.170. This does not imply that the subgroups are identical across all behavioral dimensions. In particular, the Time distance is consistently very high, approximately equal to 1, indicating strong temporal separation between subgroups. In contrast, TX, Fund, and Variant distances are very small. Consequently, the composite metric reflects that the subgroups share highly similar behavioral and structural characteristics while remaining distinguishable primarily through temporal activity.

Cluster 2

Table 6.6: Cluster 2: Average subgroup distances across behavioral and Variant dimensions.

Subgroup	Time	TX	Fund	Variant	Composite
0	1.000	0.250	0.500	0.262	0.4224
1	0.983	0.250	0.625	0.255	0.4374
2	0.983	0.250	0.625	0.255	0.4374
3	1.000	0.250	0.625	0.327	0.4758
4	0.843	0.250	0.500	0.262	0.3963
5	0.843	0.500	0.500	0.483	0.5485
6	1.000	0.250	0.625	0.255	0.4402
7	1.000	0.250	0.500	0.262	0.4224
8	0.936	0.500	0.500	0.483	0.5640

Cluster 2 exhibits moderate subgroup separation, with composite distances ranging from 0.3963 to 0.5640. Time is the dominant source of separation, with distances between 0.843 and 1.000. TX and Fund contribute more moderately, while Variant distances become particularly relevant for subgroups 5 and 8, which also exhibit the highest composite distances. These results indicate that the subgroups share some behavioral characteristics but remain distinguishable through temporal activity and process structure.

Cluster 3

Table 6.7: Cluster 3: Average subgroup distances across behavioral and Variant dimensions.

Subgroup	Time	TX	Fund	Variant	Composite
0	1.000	0.138	0.500	0.182	0.364
1	1.000	0.225	0.600	0.221	0.415
2	1.000	0.138	0.600	0.245	0.412
3	0.900	0.588	0.500	0.558	0.610
4	1.000	0.225	0.600	0.221	0.415
5	0.900	0.138	0.500	0.182	0.347
6	1.000	0.138	0.500	0.182	0.364
7	1.000	0.138	0.600	0.245	0.412
8	1.000	0.138	0.500	0.182	0.364
9	1.000	0.225	0.600	0.221	0.415
10	1.000	0.138	0.500	0.182	0.364

Cluster 3 contains the most heterogeneous set of subgroups, with composite distances ranging from 0.347 to 0.610. Subgroup 3 has the highest composite distance (0.610), driven by relatively high TX and Variant distances in addition to substantial temporal separation. In contrast, subgroup 5 has the lowest composite distance (0.347). The results suggest that temporal separation is a consistent source of distinction within the cluster, while differences in TX and Variant structure further amplify separation for particular subgroups.

6.3.3 Interpretation of Subgroup Separation

The subgroup separation analysis complements the cohesion analysis by examining whether subgroups within the same cluster remain distinguishable despite exhibiting strong internal behavioral similarity. The Composite column integrates Time, TX, Fund, and process Variant dimensions into a single measure, where larger values indicate greater average separation from the other subgroups.

Cluster 0. Cluster 0 demonstrates substantial heterogeneity across its subgroups. Subgroup 1 has the highest composite distance (0.778), primarily due to its high Time, TX, Fund, and Variant distances. This indicates that subgroup 1 follows a behavioral and structural pattern that is comparatively distinct from the remaining subgroups. Subgroups 2 and 3 exhibit lower composite distances, suggesting greater similarity to the other subgroups within the cluster.

Cluster 1. Cluster 1 exhibits the lowest composite distances, with values concentrated around 0.17. The particularly high Time distances, combined with almost zero TX and Variant distances, indicate that the subgroups are structurally and transactionally similar but differ strongly in when activity occurs. This demonstrates that temporal behavior provides an additional source of subgroup discrimination even when other dimensions are highly consistent.

Cluster 2. Cluster 2 shows moderate separation, with composite distances between 0.3963 and 0.5640. Time is consistently the strongest source of separation, while TX and Fund contribute

more moderately. Subgroups 5 and 8 show the highest composite distances and also have elevated TX and Variant distances. This suggests that these subgroups differ from the rest of the cluster not only temporally but also in their transaction intensity and process structure.

Cluster 3. Cluster 3 exhibits the widest range of composite distances, indicating greater heterogeneity among its subgroups. Subgroup 3 has the highest composite distance (0.610), whereas subgroup 5 has the lowest (0.347). Temporal separation remains substantial throughout the cluster, while TX and Variant differences distinguish particular subgroups more strongly. Thus, Cluster 3 contains both relatively similar subgroups and subgroups with pronounced behavioral and structural differences.

6.3.4 Overall Interpretation of Cluster Quality

Taken together, the cohesion and separation results provide complementary evidence for the quality of the identified Sybil subgroups. The high mean cohesion of 0.9748 indicates that wallets within individual subgroups are strongly behaviorally consistent. At the same time, the subgroup separation analysis shows that these coherent subgroups are not necessarily identical to one another.

The composite separation values reveal meaningful variation between subgroups across clusters. In particular, temporal behavior consistently provides an important source of separation, while transaction activity, fund-related behavior, and process Variant differences further distinguish specific subgroups. This is especially evident in Clusters 0, 2, and 3, where higher composite distances correspond to subgroups exhibiting stronger differences across multiple dimensions.

Therefore, the combination of high intra-subgroup cohesion and non-zero inter-subgroup separation suggests that the clustering framework produces groups that are internally coherent while retaining meaningful behavioral distinctions between subgroups. This supports the use of the proposed similarity framework for identifying distinct Sybil behavioral patterns without collapsing all related wallets into a single homogeneous group.

6.4 Evaluation of Contamination Inference

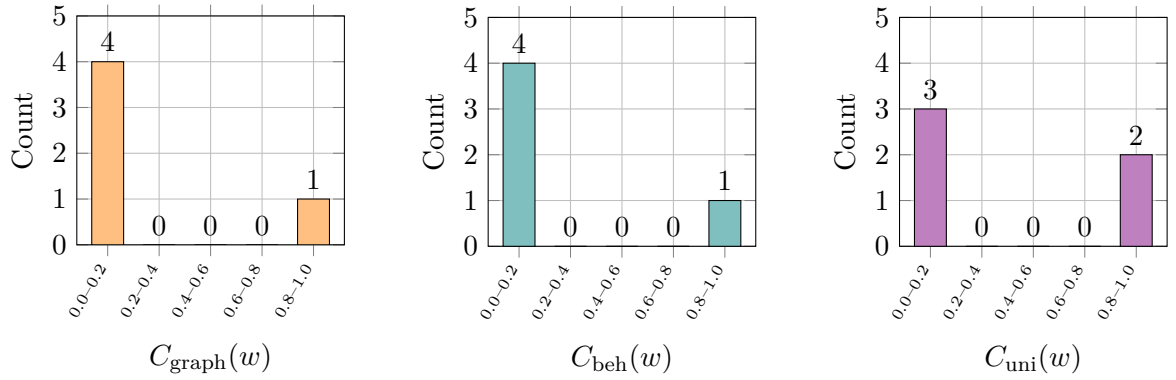
Evaluation scope and rationale. The contamination inference module is not designed as a predictive binary classifier that assigns wallets to Sybil or benign classes. Instead, it quantifies the strength and source of observable evidence that a wallet is affiliated with an independently identified Sybil cluster. Consequently, the evaluation does not rely on conventional classification metrics such as AUC-ROC or PR-AUC. Instead, we examine the behavior of the contamination scores from three complementary perspectives:

- **Score distribution analysis:** the distributions of C_{graph}, C_{beh}, and C_{uni} are examined across cases/airdrop campaigns to characterize the strength of the available evidence and to determine whether graph and behavioral views provide complementary signals. The corresponding score distributions and evidence categories are summarized in Table 5.1 and the case-level figures below.
- **Low-evidence wallet analysis:** independently verified Sybil-affiliated wallets receiving low or zero contamination scores are examined to identify cases where the available structural and behavioral observations are insufficient to produce a strong affiliation signal.
- **Coverage of verified Sybil wallets:** the proportion of independently verified Sybil-affiliated wallets receiving a high contamination score, $C(w) \geq \tau$, is reported as the high-confidence sybil affiliation rate. This measures how frequently the framework produces strong observable evidence for wallets already known to belong to Sybil wallets, rather than measuring predictive classification performance.

Given the limited sample size of 32 wallets across four sub-clusters, results are reported at both the case and aggregate levels. Complete per-wallet scores and evidence attribution are provided in the Appendix tables referenced in Section 5.3.1.

Score distribution analysis

Case 1: IRYS (ETH). Figure 6.1 shows the score distributions across the graph, behavioral, and unified views for the five wallets in the IRYS case.



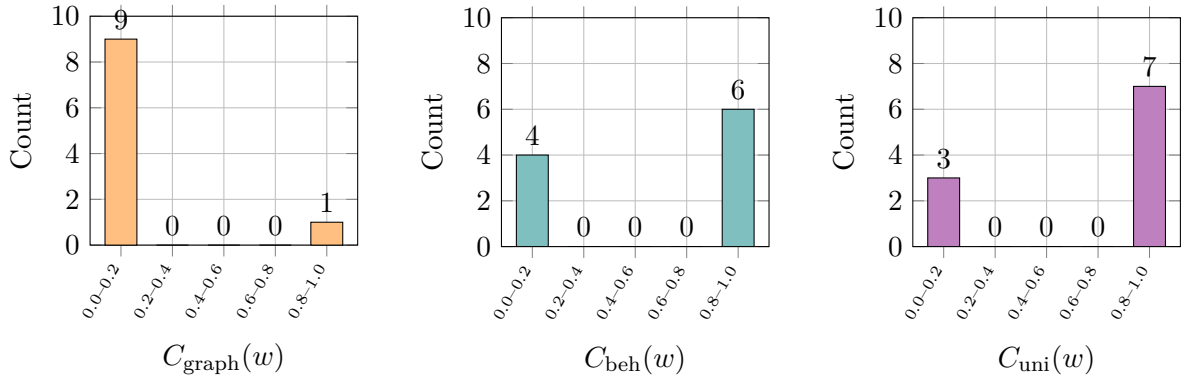
(a) Graph (b) Behavior (c) Unified

Figure 6.1: Distribution of contamination scores across the five wallets in Case 1 (IRYS), shown separately for the graph, behavioral, and unified views.

Both C_{graph} and C_{beh} exhibit a strongly bimodal distribution, with four wallets receiving scores in the $[0, 0.2)$ range and one wallet receiving a score in the $[0.8, 1.0]$ range. However, the two views assign their highest scores to different wallets. The graph view identifies one wallet through structural evidence, while the behavior view identifies another through behavioral similarity. Thus, the similar marginal distributions do not imply redundant evidence at the wallet level.

The unified view produces two high-scoring wallets, corresponding to the wallets identified by the two individual views. In this case, the combination therefore captures complementary evidence distributed across the graph and behavioral views: neither individual view alone identifies both wallets, whereas the unified view flags both as having strong observable affiliation with the verified Sybil cluster.

Case 2: MYX (BSC). Figure 6.2 shows the contamination score distributions across the graph, behavioral, and unified views for the ten wallets in the MYX case.



(a) Graph (b) Behavior (c) Unified

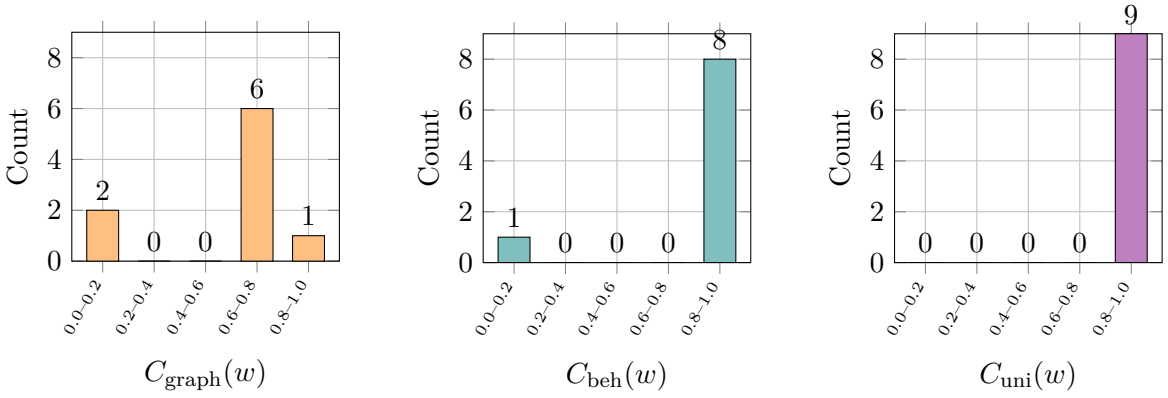
Figure 6.2: Distribution of contamination scores across the ten wallets in Case 2 (MYX), shown separately for the graph, behavioral, and unified views.

The graph view provides strong structural evidence for only one of the ten wallets, with the remaining nine receiving scores in the $[0, 0.2)$ range. In contrast, the behavioral view assigns high scores ($C_{\text{beh}} \geq 0.8$) to six wallets, indicating that behavioral similarity provides substantially more evidence of Sybil-cluster affiliation in this case than the observed graph structure.

The unified view produces seven high-scoring wallets, compared with six in the behavioral view and one in the graph view. This shows that C_{uni} is able to jointly incorporate evidence from both views: while behavioral evidence accounts for most of the high scores, the additional graph signal contributes to the unified assessment of a wallet that is not strongly identified by behavior alone. Thus, the unified score preserves the dominant behavioral signal while also incorporating complementary structural evidence. The three wallets remaining in the low-score range in the unified view receive little or no observable evidence from either channel. Overall, this case demonstrates that the relative contribution of the two views can vary substantially across Sybil clusters, with behavioral evidence being particularly informative for the MYX cluster.

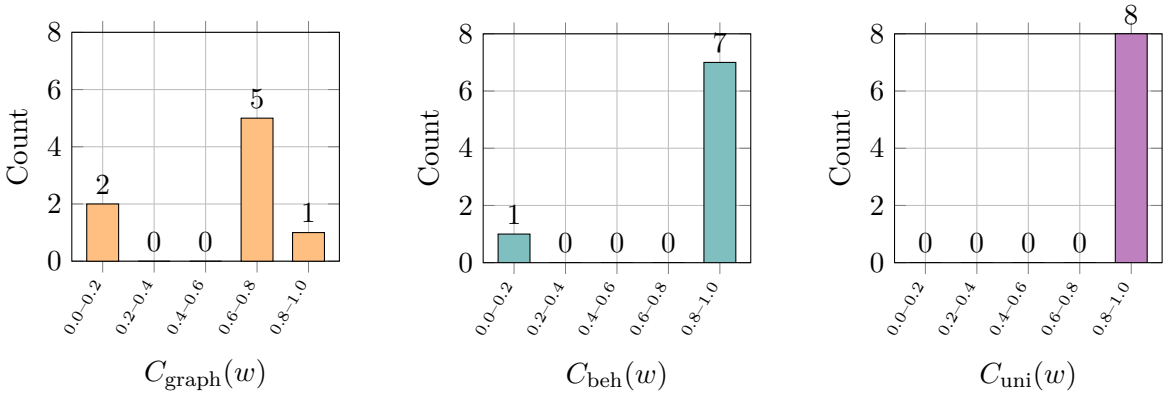
Case 3: Apriori (BSC), Clusters 1 & 2. In both Apriori clusters, the graph view shows substantial structural evidence for most wallets: six of nine wallets in Cluster 1 and five of eight wallets in Cluster 2 fall in the $[0.6, 0.8)$ range, with one additional wallet in each cluster reaching the $[0.8, 1.0]$ range. In contrast, the behavior view assigns very high scores to nearly all wallets, with eight of nine wallets in Cluster 1 and seven of eight wallets in Cluster 2 falling in the $[0.8, 1.0]$ range. Thus, both views provide evidence of cluster affiliation, although the behavioral signal is generally stronger.

The unified view jointly captures these two sources of evidence. In both clusters, every wallet receives a unified score in the $[0.8, 1.0]$ range, including wallets that do not receive a high score from one of the individual views. This indicates that the unified score combines moderate structural evidence with strong behavioral evidence rather than requiring both views to independently produce high scores. Unlike Cases 1 and 2, where one view contains several low-score wallets, Case 3 shows consistent evidence across both channels, resulting in uniformly high unified scores.



(a) Graph (Cluster 1) (b) Behavior (Cluster 1) (c) Unified (Cluster 1)

Figure 6.3: Score distributions across the three views for the nine wallets in Apriori Cluster 1.



(a) Graph (Cluster 2) (b) Behavior (Cluster 2) (c) Unified (Cluster 2)

Figure 6.4: Score distributions across the three views for the eight wallets in Apriori Cluster 2.

Ablation: contribution of individual views

The per-case breakdown above doubles as an ablation of the two individual views against the unified score:

- **Graph view in isolation:** In Cases 1 and 2, C_{graph} alone identifies only 1 wallet each as high-risk (0x1ab4...8f23 and 0xf598...b115 respectively) and returns 0 for the remaining wallets, including several with confirmed Sybil labels and strong behavioral evidence. Relying on the graph view alone in these cases would therefore miss the majority of interpretable Sybil noise.
- **Behavior view in isolation:** Conversely, in Case 1 the behavior view misses the wallet interpreted by the graph view (0x1ab4...8f23, $C_{\text{beh}} = 0$), and in both Apriori clusters the behavior view alone would still miss the two purely graph-dominant wallets (0x1fbe...508b, 0x01c9...0bb0, both $C_{\text{beh}} = 0$).
- **Necessity of the unified score:** Wallet 0x2961...b1c2 (Case 3, Cluster 2) illustrates the added value of C_{uni} beyond simply taking the maximum of the two individual views: with $C_{\text{graph}} = 0.6321$ and $C_{\text{beh}} = 0.9702$, neither view alone exceeds 0.98, yet the aggregated exposure $x(w) = x_{\text{graph}}(w) + x_{\text{beh}}(w)$ yields $C_{\text{uni}}(w) = 0.9890$ – higher than either individual view – reflecting the compounding effect of two simultaneously present, moderately strong signals. This is precisely the scenario the unified score was designed to capture: evidence that is individually inconclusive but jointly corroborating.

These observations collectively support the ensemble design of $C(w) = \max(C_{\text{graph}}, C_{\text{beh}}, C_{\text{uni}})$: no single view is sufficient across all three campaigns, and the unified score provides additional resolution precisely in the combined-signal cases where individual views under-represent the cumulative strength of the evidence.

Low-evidence wallets analysis

The remaining 6/32 wallets receive $C(w) = 0$: three in Case 1 (0x0db7...d2ca, 0x23b4...9fec, 0x8afa...734e) and three in Case 2 (0x1b70...2620, 0x6505...ffa5, 0xf2c9...f2d5). All six occur exclusively within the two star-topology cases, with none appearing in the tree-structured Apriori clusters. For these wallets, neither the K-hop structural neighborhood nor the peer behavioral similarity exceeds the respective detection thresholds, despite their confirmed Sybil-cluster membership.

One possible explanation is that the higher-level funding wallets in these star-shaped structures exhibit behavioral characteristics that differ from those of the downstream sub-wallets. In particular, root wallets that distribute funds to multiple sub-wallets tend to have higher transaction counts, different first-funding times, and different funding amounts than the downstream wallets. Their behavioral signatures are therefore distinct from those of the other sub-wallets, which can reduce the behavioral similarity measured among the latter. This provides a possible explanation for why some confirmed Sybil wallets receive little behavioral evidence despite belonging to the same underlying cluster.

The topology provides a complementary explanation for the lack of graph evidence. Cases 1 and 2 exhibit a star-like, single-layer funding pattern, in which a root wallet distributes funds directly to multiple sub-wallets without pairwise transfers among those sub-wallets. Consequently, once the root is excluded, the individual sub-wallets appear largely independent of one another in the observed funding graph. A sub-wallet may therefore be structurally connected to the funding source while remaining disconnected from its peers, limiting the graph-based evidence available for identifying shared Sybil affiliation.

Case 3, in contrast, follows a multi-layer tree-like funding structure, where funds propagate through intermediate wallets across multiple tiers. This creates additional opportunities for wallets within the same Sybil organization to become structurally connected through shared ancestors and intermediate funding paths. Structural proximity and behavioral similarity are therefore more likely to co-occur for the same wallets.

Coverage of verified Sybil wallets

Since all 32 noise wallets carry an independently established Sybil label, we evaluate how consistently the contamination score identifies these wallets as high-risk. We define the coverage rate over the verified Sybil wallets as

$$\text{Coverage}_{\text{Sybil}}(\tau) = \frac{|\{w \in \mathcal{W}_{\text{Sybil}} : C(w) \geq \tau\}|}{|\mathcal{W}_{\text{Sybil}}|}, \tag{6.6}$$

where $\mathcal{W}_{\text{Sybil}}$ denotes the set of independently verified Sybil-affiliated wallets where $\mathcal{W}_{\text{Sybil}}$ denotes the set of independently verified Sybil-affiliated wallets and τ is the high-score threshold, set to $\tau = 0.8$ in this evaluation. This is equivalent to recall with respect to the verified Sybil wallets, but is reported as a coverage rate because the evaluation does not assess classification performance over unlabeled or benign wallets. In addition to the overall coverage, we examine which evidentiary dimension contributes to the high score in each case:

- **Case 1 (IRYS):** 2/5 wallets (40%) receive a contamination score above the threshold, with one identified primarily through graph exposure and the other through behavioral similarity. This reflects the limited overlap between the two evidentiary channels in the star-shaped funding structure.
- **Case 2 (MYX):** 7/10 wallets (70%) receive a contamination score above the threshold, with behavioral similarity providing the dominant source of evidence and graph exposure contributing to only a small number of wallets.
- **Case 3 (Apriori, both clusters):** 17/17 wallets (100%) receive a unified score above the threshold, with $C(w) \geq 0.965$. In this case, the unified score captures the combination of moderate structural exposure ($C_{\text{graph}} \approx 0.63$) and strong behavioral similarity ($C_{\text{beh}} \geq 0.96$).

Overall, 26/32 verified Sybil wallets (81%) receive a contamination score above the evaluation threshold, $C(w) \geq \tau = 0.8$. For these wallets, the framework also provides an interpretable account of the evidence underlying the score by separating structural and behavioral contributions. The remaining six wallets are examined separately in the low-evidence wallet analysis above.

6.5 Non-Sybil Control Wallets

The control-wallet experiment produced no observed false positives: none of the 21 non-Sybil control wallets exceeded the evaluation threshold. This indicates that, within the evaluated dataset, the framework did not assign contamination evidence to wallets that were neither part of the known Sybil clusters nor connected to them through the observed funding graph. However, this result should not be interpreted as a general false-positive rate of zero, since the control set is limited and does not represent the full population of non-Sybil wallets.

6.6 Sensitivity & Robustness

6.6.1 Robustness and Sensitivity

DBSCAN `eps` Sensitivity We evaluate the sensitivity of the primary clustering stage to the DBSCAN neighborhood parameter `eps`. The analysis examines how changes in `eps` affect three properties of the resulting clustering structure:

- **Mean cohesion:** the average pairwise similarity within detected subgroups, measuring the overall compactness of the resulting behavioral groups;
- **Cohesion variance:** the variance of cohesion values across subgroups, indicating the consistency of subgroup quality;
- **Total noise count:** the number of wallets not assigned to a DBSCAN cluster, reflecting the strictness of the density requirement and determining the set of wallets subsequently considered by the contamination inference module.

Lower values of `eps` impose a stricter neighborhood requirement and therefore tend to preserve only highly cohesive subgroups, while increasing the number of wallets classified as noise. Increasing `eps` relaxes the density requirement, allowing more wallets to be incorporated into clusters but potentially reducing subgroup cohesion. The parameter therefore represents a trade-off between subgroup compactness, clustering resolution, and the amount of unresolved activity passed to the subsequent contamination analysis.

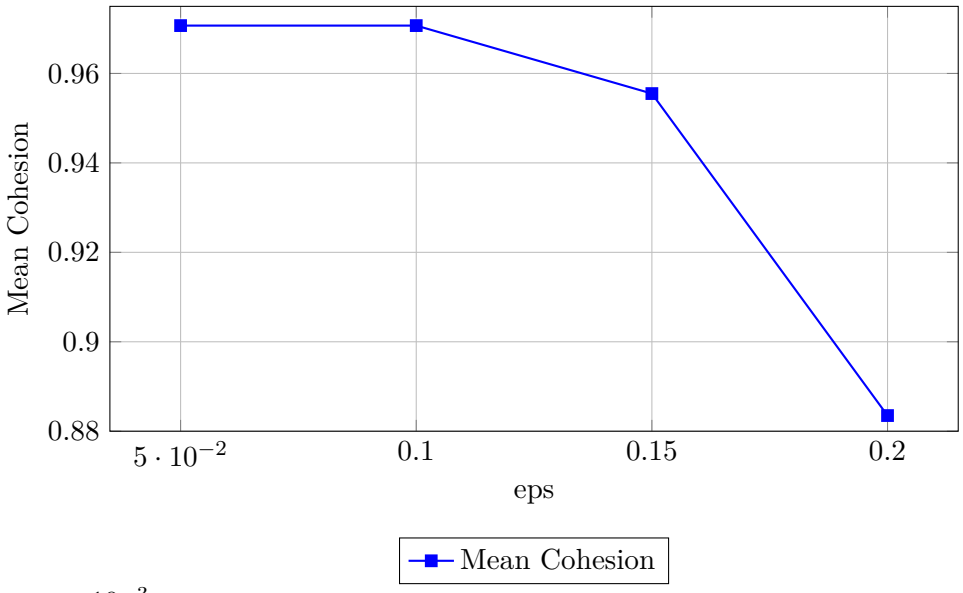
Choice of `eps`=0.1–0.15 The results show that both `eps`=0.1 and `eps`=0.15 operate within a stable region of the clustering process. In particular, `eps`=0.1 achieves slightly higher mean cohesion and lower cohesion variance, while producing a more fine-grained subgroup structure and a somewhat larger noise set. By contrast, `eps`=0.15 produces fewer and less fragmented subgroups and a smaller noise set, at the cost of a modest reduction in cohesion.

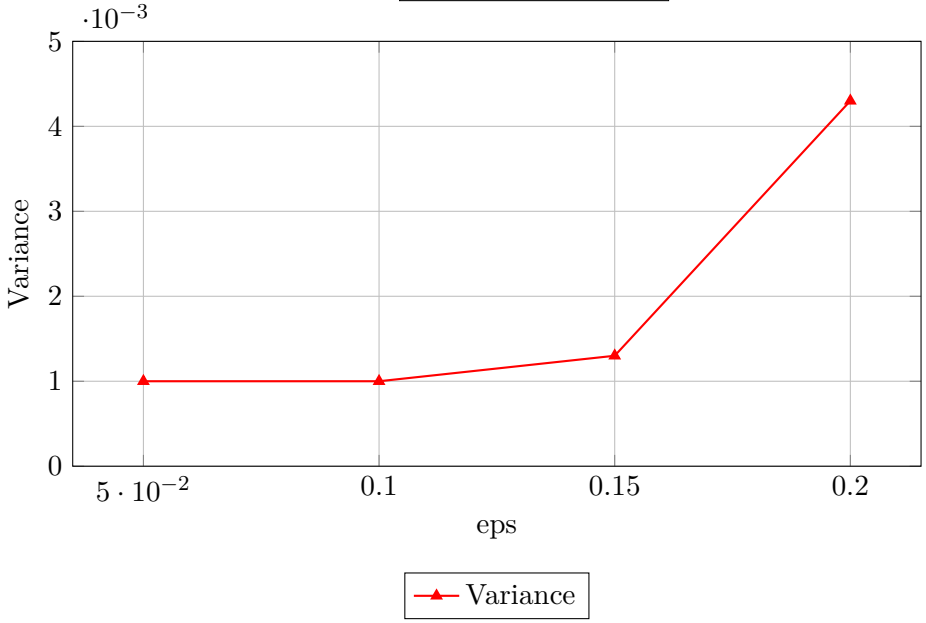
Given this trade-off, both settings provide plausible operating points for the proposed framework. We select `eps`=0.1 as the primary configuration for downstream analysis because its higher resolution preserves finer-grained behavioral coordination patterns and provides a more conservative clustering boundary. The `eps`=0.15 configuration is retained as a nearby alternative representing a slightly less granular clustering regime with fewer unresolved wallets. The relative stability of the results across these neighboring settings supports the use of `eps`=0.1 as the primary configuration without implying that it is a uniquely optimal parameter value.

Evaluation of Cohesion Across eps Values

Table 6.8: Mean Cohesion, Variance, and Total Noise for Different eps Values

eps	Total Subgroups	Mean Cohesion	Variance	Total Noise
0.2	15	0.8836	0.0043	6
0.15	28	0.9586	0.0013	26
0.1	31	0.9748	0.0010	32
0.05	31	0.9748	0.0010	39





6.6.2 Adversarial Temporal Perturbation

To examine the sensitivity of the proposed framework to a simple form of behavioral evasion, controlled perturbations are introduced to the first funding timestamps of the evaluated Sybil wallets. This models an adversary that attempts to weaken short-term temporal coordination while leaving transaction amounts, transaction counts, funding relationships, and process variants unchanged. For each perturbation level $p \in \{5, 10, 15\}$ minutes, the original timestamp t_i is replaced by

$$t'_i = t_i + \delta_i, \qquad \delta_i \sim U(-p, p). \tag{6.7}$$

The perturbed datasets are processed using the same pipeline and configuration as the original evaluation. To capture both the effect on primary Sybil verification and the underlying behavioral structure, we report the verification rate, the resulting number of DBSCAN noise wallets, and the mean subgroup cohesion.

Table 6.9: Sybil verification under ±5 minute temporal perturbation (low turbulence).

Project	Total Wallets	Verified Sybil	Noise	Verification Rate
MYX (BSC)	151	133	18	88.1%
Apriori Tree 1 (BSC)	55	42	13	76.4%
Apriori Tree 2 (BSC)	49	35	14	71.4%
IRYS (ETH)	152	141	11	92.8%
Overall	**407**	**351**	**56**	**86.2%**

Table 6.10: Sybil verification under ±10 minute temporal perturbation (medium turbulence).

Project	Total Wallets	Verified Sybil	Noise	Verification Rate
MYX (BSC)	151	122	29	80.8%
Apriori Tree 1 (BSC)	55	35	20	63.6%
Apriori Tree 2 (BSC)	49	28	21	57.1%
IRYS (ETH)	152	134	18	88.2%
Overall	**407**	**319**	**88**	**78.4%**

Table 6.11: Sybil verification under ±15 minute temporal perturbation (high turbulence).

Project	Total Wallets	Verified Sybil	Noise	Verification Rate
MYX (BSC)	151	111	40	73.5%
Apriori Tree 1 (BSC)	55	29	26	52.7%
Apriori Tree 2 (BSC)	49	22	27	44.9%
IRYS (ETH)	152	127	25	83.6%
Overall	**407**	**289**	**118**	**71.0%**

The verification results show a clear and monotonic degradation as temporal perturbation increases. The overall Sybil verification rate decreases from the original 92.1% to 86.2% under a ±5 minute perturbation, 78.4% under ±10 minutes, and 71.0% under ±15 minutes. At the same time, the number of wallets classified as noise increases from 32 in the original setting to 56, 88, and 118, respectively. The effect is observable across all four evaluated project structures,

although the magnitude of degradation differs between projects.

In particular, the Apriori tree-based clusters show a stronger reduction in verification coverage than the IRYS cluster. Under a ±15 minute perturbation, verification decreases to 52.7% and 44.9% for the two Apriori clusters, compared with 83.6% for IRYS. This suggests that the sensitivity of the framework is influenced not only by the magnitude of temporal perturbation but also by the underlying coordination structure and behavioral organization of the campaign.

The corresponding changes in subgroup cohesion are summarized below.

Table 6.12: Mean subgroup cohesion under increasing temporal perturbation.

Cluster	**Original**	±5 **min**	±10 **min**	±15 **min**
Cluster 0	0.9395	0.8268	0.7047	0.6389
Cluster 1	0.9892	0.8705	0.7419	0.6727
Cluster 2	0.9760	0.8589	0.7320	0.6637
Cluster 3	0.9784	0.8610	0.7338	0.6653
Mean	**0.9748**	**0.8543**	**0.7281**	**0.6602**

The mean subgroup cohesion decreases from 0.9708 in the original setting to 0.8543 under a ±5 minute perturbation, 0.7281 under ±10 minutes, and 0.6602 under ±15 minutes. These correspond to reductions of approximately 12%, 25%, and 32%, respectively. The same downward trend is observed across all evaluated clusters, indicating that the effect is not confined to a single cluster.

The joint movement of verification rate, noise count, and subgroup cohesion provides evidence that temporal perturbation affects the primary verification stage by weakening the behavioral similarity structure used for density-based clustering. However, the degradation does not result in a complete loss of cluster structure. This is consistent with the multi-dimensional design of the proposed similarity mechanism: temporal coordination represents only one component of the behavioral representation, while transaction intensity, funding characteristics, and process variants provide additional similarity signals. Perturbing the first funding timestamps therefore weakens one source of evidence without removing the remaining behavioral information.

The experiment consequently provides an empirical characterization of the framework's sensitivity to temporal obfuscation. Mild perturbation reduces verification coverage while preserving substantial behavioral structure, whereas stronger perturbations produce progressively larger degradation. This behavior is consistent with the intended scope of the framework, which focuses on the verification of strongly coordinated Sybil activity whose observable coordination patterns provide the basis for detection. Deliberately disrupting these patterns is therefore expected to reduce verification performance as less observable evidence remains available. The results should consequently be interpreted as defining the boundary of the framework's applicability under increasing behavioral obfuscation rather than as evidence of a failure of the proposed approach. More extensive adversarial strategies that simultaneously modify funding amounts, transaction intensity, process variants, or funding structures may produce a stronger degradation and remain outside the scope of the present experiment.

6.7 Supervised Baseline Comparison

To provide a conventional predictive reference point, two supervised gradient-boosted tree models, XGBoost and LightGBM, are evaluated using the same wallet-level behavioral features, labels, preprocessing, and train–test split as the proposed framework. The purpose of this comparison is not to treat the proposed framework as a conventional binary classifier, but to establish how strongly standard supervised models can exploit the available behavioral signals on the same dataset.

Both baseline models are evaluated without dataset-specific hyperparameter tuning. The XGBoost classifier uses its standard configuration with the class-imbalance correction `scale_pos_weight`, computed from the class distribution of the training set. LightGBM is evaluated using its standard configuration without additional hyperparameter optimization. This keeps the baseline comparison lightweight and reduces the influence of model-specific tuning choices.

Table 6.13: Performance of supervised baseline models on the evaluated wallet dataset.

Model	**Accuracy**	**Precision**	**Recall**	**F1**
LightGBM	89.5%	1.00	0.89	0.94
XGBoost	90.5%	1.00	0.90	0.95

Both supervised baselines achieve strong predictive performance on the evaluated dataset. XGBoost achieves an accuracy of 90.5% and a Sybil-class F1 score of 0.95, while LightGBM achieves 89.5% accuracy and a Sybil-class F1 score of 0.94. Both models achieve perfect precision for the Sybil class on the evaluated test set, while recall remains slightly below 0.90. These results indicate that the engineered behavioral features contain substantial predictive information for distinguishing the evaluated Sybil and control wallets. We note that the underlying dataset is strongly imbalanced (407 Sybil-labeled versus 21 control wallets), so the test-set negative class contains only a small number of samples. As a result, the reported precision is sensitive to individual misclassifications and should be interpreted with this sample-size limitation in mind, rather than as a tightly calibrated estimate.

The feature-importance profiles of the two models are also highly consistent. The features used by the supervised baselines capture different aspects of local wallet coordination. `time_neighbors_5m` counts the number of reference wallets whose first funding timestamps fall within a five-minute window of the target wallet, capturing short-term temporal synchronization. `amount_neighbors` counts wallets whose first funding amounts differ from that of the target wallet by no more than 20%, capturing similarity in funding magnitude. `local_density` measures the aggregate local similarity of wallets based on both funding time and funding amount, with temporal proximity receiving a decaying weight. Finally, `nn_distance` measures the standardized Euclidean distance from the target wallet to its nearest neighbor in the first-funding-time and first-funding-amount space.

Both models place `time_neighbors_5m` and `local_density` clearly above the remaining features. Together, these two features account for approximately 89% of the reported tree-based feature importance in both models, indicating that short-term temporal concentration and local behavioral density are the dominant predictive signals in the evaluated dataset. The close

Table 6.14: Feature importance of the supervised baseline models.

Feature	XGBoost	LightGBM
`time_neighbors_5m`	0.5759	0.5670
`local_density`	0.3147	0.3215
`amount_neighbors`	0.0667	0.0681
`nn_distance`	0.0426	0.0434

agreement between XGBoost and LightGBM also suggests that these signals are not specific to a particular tree-based model.

The proposed framework is evaluated using different, task-specific measures. Its primary clustering stage achieves an overall Sybil verification rate of 92.1% (375/407), while none of the 21 verified control wallets is incorrectly included in a verified Sybil cluster, corresponding to an observed false-positive rate of $0/21 = 0\%$. In addition, the contamination inference module assigns a high score $C(w) \geq 0.8$ to 26 of 32 independently verified Sybil noise wallets (81%). These metrics are not directly comparable to the classification metrics reported for XGBoost and LightGBM; rather, they evaluate the verification, control-wallet separation, and noise-interpretation capabilities specific to the proposed framework.

The comparison therefore serves as a predictive reference rather than a direct ranking of the three approaches. The strong and highly consistent performance of XGBoost and LightGBM provides empirical evidence that the behavioral features used in this study contain meaningful predictive signals for coordinated Sybil activity. In particular, the dominance of `time_neighbors_5m` and `local_density` across both models indicates that short-term temporal coordination and local behavioral density are important characteristics of the evaluated Sybil activity.

The proposed framework builds on the same observable behavioral signals, but uses them to provide a different and more directly interpretable form of verification. Rather than reducing these signals to a predictive class label, it explicitly represents wallet-to-wallet behavioral similarity and combines this evidence with traceable funding relationships. For an individual wallet, the resulting assessment can therefore be linked to specific behavioral dimensions, funding paths, graph proximity, and the corresponding structural or behavioral contamination evidence. On the evaluated data, the proposed framework achieves a verification rate within a similar numerical range to the predictive performance of the supervised baselines, while exposing a richer and more directly verifiable evidence trail. The baseline comparison therefore highlights a distinction between predictive feature importance and relationship-level interpretability: the supervised models show which behavioral signals are predictive, whereas the proposed framework makes the behavioral and funding relationships underlying an individual wallet assessment directly inspectable.

6.8 Chapter Summary

This chapter provides the empirical evaluation of the proposed Sybil verification pipeline across real-world blockchain campaigns. The evaluation examines the extent to which the proposed behavioral and funding-based model captures coordinated wallet activity, the effectiveness of the selected features and similarity measures, the ability of contamination inference to interpret

DBSCAN noise, and the robustness of the pipeline under different clustering parameters and adversarial conditions. In addition, XGBoost and LightGBM are used as supervised predictive baselines to provide a reference point for the predictive strength of the engineered behavioral features.

The results provide empirical evidence for the behavioral and funding-based characterization of potential Sybil activity developed earlier in the thesis – speaking to the modeling question of *what criteria can identify potential Sybil activity*. Across the evaluated campaigns, the proposed representation and clustering process recover highly coordinated Sybil subgroups with consistently high verification rates and strong subgroup cohesion, while the control-wallet evaluation shows no observed false positives on the evaluated benign set.

The supervised baseline results further show that conventional tree-based classifiers can achieve strong predictive performance using the same behavioral information. Notably, both XGBoost and LightGBM independently identify short-term temporal proximity and local behavioral density as their most important features, despite being trained without any knowledge of the proposed similarity design – an external, model-agnostic confirmation, on the question of *which features and similarity measures are effective*, that the criteria underlying the Weighted Decay Mechanism are genuinely informative rather than an arbitrary design choice. The comparison therefore serves as a predictive reference for contextualizing the proposed framework rather than a direct ranking against its task-specific verification measures.

The contamination-inference experiments further show that wallets excluded by the primary clustering stage may still contain meaningful structural or behavioral evidence of Sybil association. Across the evaluated noise wallets, the complementary graph and behavioral views reveal different evidence patterns across campaigns, supporting the use of multi-view contamination analysis rather than treating all noise as irrelevant.

Finally, the sensitivity analysis demonstrates that clustering quality and the size of the noise set depend on the selected DBSCAN configuration. The adversarial temporal perturbation experiment extends this into the question of *robustness and adversarial vulnerability*, measuring how verification rate and subgroup cohesion degrade under increasing timestamp perturbation (± 5 to ± 15 minutes): both decline steadily rather than collapsing outright, with the tree-structured Apriori clusters proving more sensitive than the star-structured IRYS cluster. This confirms that the framework relies on short-term temporal synchronization as intended – an adversary randomizing this signal is moving outside the star/tree coordination scope defined in Section 1.3, rather than defeating a mechanism claimed to cover it. Taken together, these results replace a purely qualitative limitation with a measured degradation curve marking the framework's operational boundary, providing the empirical basis for the discussion of contributions and limitations in the following chapter.

Chapter 7

Discussion

7.1 Overall Discussion

Our evaluation demonstrates that the proposed method effectively recovers highly coordinated Sybil wallets with high precision. The DBSCAN clustering combined with the Weighted Decay Mechanism achieves strong performance in identifying clustered Sybil groups, while the contamination inference module extends this analysis to previously unassigned noise wallets.

Instead of discarding DBSCAN noise, the contamination inference module re-evaluates these wallets using graph-based exposure and behavioral similarity to known Sybil structures. Across the three case studies analyzed in Section 6.4, 26 of 32 noise wallets with independently confirmed Sybil-cluster membership (81%) received a high contamination score ($C(w) \geq 0.8$). In every such case, the evidentiary dimension driving the score – structural exposure, behavioral similarity, or their combination – could be explicitly traced. This indicates that a substantial portion of the noise wallets carry meaningful and interpretable evidence of Sybil association that is not captured by the primary clustering stage alone. We note that this evaluation is recall-oriented: because all analyzed noise wallets in this dataset carry a confirmed Sybil label, we are not yet able to quantify the module's false positive behavior on confirmed-benign noise wallets. This remains an open point for future evaluation, as discussed in Section 7.4.

These findings provide empirical support for the characterization of Sybil behavior considered in **RQ2**. Across the evaluated campaigns, coordinated activity is reflected through both behavioral consistency within wallet subgroups and funding relationships that connect otherwise distinct groups. At the same time, the results show that these two forms of evidence do not contribute uniformly across campaigns: their relative strength depends on the coordination topology and behavioral pattern exhibited by the underlying activity.

Cluster cohesion remains consistently high, while subgroup separation distinguishes behavioral patterns across the Time, TX, Fund, and Variant dimensions. This indicates that the Weighted Decay Mechanism preserves meaningful structure under heterogeneous behavioral conditions. The results therefore provide empirical support for **RQ3**, showing that effective wallet similarity depends not only on the choice of similarity measure, but also on how behavioral intensity, temporal structure, and process-level information are represented and combined.

The adversarial temporal perturbation experiment (6.6.2) extends this discussion to **RQ4**. Rather than treating adversarial robustness as a purely theoretical concern, the experiment

quantifies how the framework degrades under a targeted evasion attempt: as timestamp perturbation increases from ±5 to ±15 minutes, the overall verification rate declines from 92.1% to 71.0% and mean subgroup cohesion drops by roughly a third, with the degradation proceeding gradually rather than collapsing outright. The tree-structured Apriori clusters prove more sensitive than the star-structured IRYS cluster, indicating that robustness is shaped by coordination topology as much as by the perturbation itself. This behavior is consistent with the scope defined for the proposed method (Section 1.3): the framework is designed to verify strongly coordinated, structurally observable star- and tree-like Sybil activity, and an adversary that deliberately randomizes funding timing is moving outside this scope rather than exposing a flaw in a mechanism that claimed to cover it. The experiment therefore does not resolve the question of adversarial robustness in general, but it replaces a purely qualitative limitation with a measured degradation curve that delineates where the framework's underlying assumptions begin to break down.

Overall, the results support a two-stage design: density-based clustering for core Sybil detection, followed by contamination inference as an interpretability layer for identifying and characterizing sparsely connected or borderline coordinated Sybil-associated wallets that the primary clustering stage leaves unresolved. At the same time, the adversarial perturbation results indicate that this design is best understood as effective within a defined operating range – strongly coordinated, temporally synchronized Sybil activity – rather than as a general-purpose solution robust to arbitrary evasion strategies.

7.2 Limitations

- **Robustness to adversarial and cross-project variation:** The proposed method primarily focuses on detecting star- and tree-like coordination structures, where Sybil wallets exhibit relatively centralized funding relationships or hierarchical propagation patterns. More advanced adversaries may deliberately avoid such structures by constructing mesh-like or highly decentralized multi-hop transaction graphs. In these cases, funds may be redistributed through multiple intermediate wallets, obscuring direct funding relationships and weakening the assumptions of first-occurrence fund tracing.

  Adversaries may also introduce randomized behavioral perturbations, including variable transaction amounts, non-deterministic timing delays, and benign-looking transactions within Sybil-related flows. These strategies reduce structural density, weaken temporal coherence, and distort behavioral similarity, making coordinated activity less distinguishable from normal behavior. Cross-chain bridges, mixing services, and dynamically rotating intermediary wallets may further fragment the observable transaction graph and limit the recovery of complete coordination structures.

  The framework is also deliberately precision-oriented. Wallets that deviate from dominant behavioral patterns may therefore be classified as noise, which reduces false positives but can lower recall for partially coordinated or weakly structured Sybil activity. More broadly, different blockchain projects exhibit heterogeneous transaction behaviors, funding mechanisms, and user interaction patterns. Consequently, fixed parameter settings

and behavioral assumptions may not transfer directly across ecosystems and may require project-specific calibration. These observations indicate that the current framework is best suited to strongly coordinated and structurally observable Sybil activity rather than weakly structured or highly obfuscated attacks.

- **Method and parameter dependence:** The performance of the framework is sensitive to hyperparameter selection, particularly the DBSCAN neighborhood radius (`eps`) and the weighting configuration in the weighted decay similarity function. Variations in these parameters can significantly affect clustering granularity and similarity sensitivity. This dependence also propagates to the contamination inference module because the set of noise wallets it analyzes is determined by the preceding DBSCAN configuration. Consequently, contamination scores should be interpreted as conditional on the selected clustering and similarity parameters rather than as parameter-independent properties of the analyzed wallets. The sensitivity of the contamination analysis to `eps` is examined empirically in Section 6.6.1.

- **Evaluation scope of contamination inference:** All 32 noise wallets analyzed in the contamination inference case studies carry an independently confirmed Sybil label, while no confirmed-benign noise wallets were available for contrast. The evaluation therefore establishes that the contamination score can assign high scores to known Sybil noise wallets with high consistency, but does not yet establish how the score behaves on noise wallets unrelated to Sybil activity. Standard discriminative metrics such as AUC-ROC or PR-AUC, which require both positive and negative classes, could not be computed for this reason. Future evaluation should include confirmed-benign noise wallets to assess the false-positive behavior and discriminative performance of the contamination inference module.

- **Class imbalance and limited negative samples:** The evaluated dataset is strongly imbalanced, comprising 407 Sybil-labeled wallets against only 21 independently verified control wallets. This imbalance limits the statistical reliability of classification-oriented metrics reported in this thesis, particularly precision and accuracy, since even a small number of misclassified control wallets would substantially shift these values. The supervised baseline comparison in Section 6.7 is affected in the same way: the small number of negative samples in the corresponding test split means the reported perfect precision should be interpreted as reflecting the limited sample size rather than a tightly calibrated estimate of false-positive behavior. A larger and more diverse set of confirmed-benign wallets, spanning multiple projects and interaction patterns, would be required to obtain more statistically robust estimates of the framework's false-positive rate and of supervised baseline precision.

7.3 Practical Implications

The proposed method provides a scalable, interpretable, and evidence-driven framework for blockchain security analysis. Beyond Sybil detection in airdrop systems, the same pipeline can

be generalized to a broader class of on-chain behavioral intelligence tasks, where coordination, funding relationships, and temporal structure are key signals. Key application scenarios include:

- **Interpretable Sybil verification in airdrop campaigns:** Identifying coordinated wallet clusters with shared behavioral and funding patterns, and providing verifiable evidence chains for enforcement and appeal processes. This serves as an additional cross-validation layer for ML/DL-based anomaly detection models, improving trust, transparency, and false positive auditing.
- **Rug pull and token manipulation analysis:** Detecting coordinated early-stage behavior such as distributed gas funding to newly created wallets followed by synchronized token sniping, enabling reconstruction of pre-launch coordination structures.
- **Prediction-market insider detection:** Detecting potentially coordinated or insider-like trading activity by analyzing a specified event and its associated markets, including batch creation of new wallets, shared funding sources, coordinated betting across related outcome options, anomalous trading behavior, and abnormal profit patterns.
- **Operational support for security teams:** Supporting evidence-based investigation through a more relationship-aware analysis of Sybil activity. In addition to behavioral similarity, the framework incorporates funding relationships and graph structure to provide causal-like evidence of how wallets may be connected to known Sybil activity, while contamination inference extends this analysis to otherwise discarded noise wallets. The resulting evidence can help analysts identify potentially related intermediary or "mule" wallets and prioritize them for further investigation, with the contamination score $C(w)$ serving as a triage signal rather than an automated reclassification mechanism.

7.4 Future Work

Building on the limitations identified in this study, several promising directions can be explored to further enhance the robustness, scalability, and generalisability of the proposed Sybil detection framework.

- **Extension to complex coordination structures:** Future work should extend the current framework to better handle increasingly sophisticated Sybil evasion strategies that go beyond structural coordination patterns. In particular, adversaries may rely on relay-based transaction routing, token swapping mechanisms, and intermediate wallet chains to obscure direct fund provenance and weaken traceability. These strategies can fragment the observable transaction graph and reduce the effectiveness of purely structural detection approaches.

  This motivates the need to incorporate more expressive representations that jointly model fund-flow transformations and behavioral dependencies, enabling the system to remain robust under obfuscation techniques that preserve functionality while disguising coordination.

- **Benign-labeled validation for contamination inference:** A direct extension of the evaluation presented in this thesis is the incorporation of confirmed-benign noise wallets, either through manual review or protocol-provided labels, to assess the false positive behavior of the contamination score. This would complement the recall-oriented evaluation performed here and allow standard discriminative metrics to be computed where currently only consistency-based analysis is possible.

- **Forensic follow-up on unexplained noise wallets:** Six of the 32 analyzed noise wallets received a contamination score of $C(w) = 0$ despite having confirmed Sybil-cluster membership. Future work should investigate these cases individually—for example, by monitoring subsequent on-chain activity for potentially dormant intermediary wallets or by extending the graph view's K-hop horizon—to determine whether the absence of contamination evidence reflects genuinely weak observable connections or limitations of the current feature scope.

- **Adaptive parameter learning and density modelling:** To reduce reliance on manual hyperparameter tuning, future research should focus on adaptive parameter selection mechanisms for both similarity weighting and clustering components. In particular, density-aware clustering approaches such as HDBSCAN should be investigated as potential replacements for DBSCAN, as they can better handle variable-density structures and improve robustness across heterogeneous blockchain datasets. Such approaches may also reduce the sensitivity of the contamination inference module's noise set to a manually chosen `eps` value.

Chapter 8

Conclusion

This thesis investigated interpretable Sybil wallet detection in blockchain airdrop campaigns, with a focus on identifying coordinated activity across multiple wallets using observable on-chain evidence. The proposed framework combines funding relationships with behavioral similarity to provide an interpretable verification layer that complements existing Sybil detection approaches. The following sections summarize the findings by directly answering the research questions, followed by the overall experimental outcome, the main contributions, and directions for future research.

8.1 Answer to Research Questions

RQ1: Performance of Existing Sybil Detection Methods

Existing Sybil detection approaches, including heuristic-based methods and graph neural networks, demonstrate strong capability in identifying anomalous wallet behavior. However, they differ substantially in their assumptions, data requirements, and interpretability. Heuristic approaches may depend on external signals such as IP addresses or off-chain metadata, which can be difficult to obtain or deliberately obfuscated, while deep learning approaches can achieve strong predictive performance without providing explicit reasoning paths that can be independently inspected or verified. These findings highlight a gap between predictive capability and forensic transparency, motivating the need for complementary approaches that provide explicit and auditable evidence.

RQ2: Modeling Sybil Behavior

The results show that Sybil behavior can be meaningfully modeled from on-chain wallet data through coordinated activity across multiple wallets. Potential Sybil activity can be characterized by two complementary forms of evidence: behavioral coordination and shared funding provenance. Behavioral coordination captures similarities in wallet activity, while shared funding provenance provides structural evidence linking otherwise seemingly unrelated wallets to a common funding source. Together, these criteria provide an interpretable basis for identifying potential Sybil activity beyond isolated wallet-level anomalies.

RQ3: Effective Features and Similarity Measures

The evaluation shows that conventional similarity measures such as Euclidean distance and cosine similarity are insufficient for fully capturing the coordinated wallet behavior observed in the analyzed campaigns. Euclidean distance is sensitive to differences in feature scale and may be affected by temporal smoothing, while cosine similarity is invariant to vector magnitude and can therefore regard wallets with different transactional intensities as similar when their feature vectors point in the same direction. In contrast, the combination of continuous transaction features and discrete behavioral sequences provides stronger discriminative power. In particular, the weighted decay mechanism, incorporating feature weighting, exponential temporal decay, and normalized Levenshtein distance, is effective for capturing synchronized multi-round behavioral patterns.

RQ4: Limitations and Adversarial Evasion

The findings indicate that the proposed framework is most effective for strongly coordinated and structurally observable Sybil activity. Its reliance on observable funding relationships, behavioral consistency, and density-based clustering creates potential weaknesses that adaptive adversaries may exploit through multi-hop or decentralized fund routing, randomized transaction behavior, noise injection, and privacy-enhancing mechanisms. These strategies can weaken funding provenance, reduce behavioral similarity, and distort cluster formation. The framework is also sensitive to clustering and similarity parameters, and the current contamination-inference evaluation is limited by the absence of confirmed-benign noise wallets. Consequently, highly adaptive and obfuscated Sybil campaigns remain challenging to characterize and detect, while broader validation is required to establish the generality and false-positive behavior of the contamination inference module.

Experimental Outcome

The proposed framework was evaluated on 428 real-world wallets across three blockchain ecosystems. It achieved a 92.1% Sybil identification rate across the evaluated campaigns, including MYX, Apriori Tree 1, Apriori Tree 2, and IRYS. On a control set of 21 verified benign wallets, the framework produced an observed false-positive rate of $0/21 = 0\%$. In addition, the contamination inference analysis assigned a high contamination score ($C(w) \geq 0.8$) to 26 of 32 independently confirmed Sybil noise wallets (81%). In these cases, the evidence contributing to the score could be traced to structural exposure, behavioral similarity, or their combination.

Taken together, these results support a two-stage design in which density-based clustering identifies strongly coordinated Sybil activity, while contamination inference provides an additional interpretability layer for evaluating sparse or borderline wallets that the primary clustering stage leaves unresolved. The framework therefore complements black-box detection approaches by providing explicit funding and behavioral evidence and by extending investigation to noise wallets that would otherwise be discarded. This also supports practical security analysis by allowing analysts to inspect the evidence underlying suspicious wallet associations and prioritize unresolved cases for further investigation.

8.2 Core Contributions

This thesis makes three primary contributions:

1. **Multi-view Sybil Detection via Funding and Behavior Integration:** The framework combines funding relationships with sequential behavioral modeling to provide complementary evidence of coordinated wallet activity. This provides an auditable evidence-based layer that complements probabilistic Sybil detection approaches.

2. **Weighted Decay Similarity Mechanism:** A similarity mechanism combining feature weighting, exponential temporal decay, and normalized Levenshtein distance is introduced to capture both transactional intensity and behavioral synchronization. The mechanism improves the representation of fine-grained, multi-round automation patterns that are not adequately captured by conventional similarity measures.

3. **Contamination Inference for Noise Interpretation:** A contamination inference mechanism is introduced to further evaluate DBSCAN noise wallets using graph-based exposure, behavioral similarity, and unified evidence. Rather than treating all noise as irrelevant, the framework provides a structured way to identify and characterize wallets that may remain associated with known Sybil activity despite falling outside the primary clusters.

8.3 Future Work

Building on the limitations discussed in Section 7.2, several directions stand out as the most immediate next steps: extending the funding model beyond first-occurrence tracing to handle more complex coordination structures, validating the contamination inference module against confirmed-benign noise wallets to enable standard discriminative metrics, and investigating adaptive clustering approaches such as HDBSCAN to reduce dependence on manually chosen parameters.

Overall, the findings suggest that interpretable Sybil detection can benefit from combining behavioral coordination with explicit funding evidence. While the proposed framework does not eliminate the challenges posed by adaptive adversaries, it demonstrates that coordinated wallet activity can be analyzed through observable and traceable evidence rather than treated solely as a black-box classification problem.

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Chapter 9

Appendix

.1 Full Noise Wallet Listings (eps=0.1)

Table 1: Case 1 (IRYS) – full listing of all 5 noise wallets

Wallet	x_{graph}	x_{beh}	C_{graph}	C_{beh}	C_{uni}	$C(w)$
0x1ab4...8f23	147.0000	0.0000	1.0000	0.0000	1.0000	1.0000
0x0db7...d2ca	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
0x23b4...9fec	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
0x0722...a9c6	0.0000	121.8004	0.0000	1.0000	1.0000	1.0000
0x8afa...734e	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Table 2: Case 2 (MYX) – full listing of all 10 noise wallets

Wallet	x_{graph}	x_{beh}	C_{graph}	C_{beh}	C_{uni}	$C(w)$
0xf598...b115	141.0000	0.0000	1.0000	0.0000	1.0000	1.0000
0x1b70...2620	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
0x9621...45b6	0.0000	9.0050	0.0000	0.9999	0.9999	0.9999
0xa05e...c970	0.0000	5.8143	0.0000	0.9970	0.9970	0.9970
0xbf65...b315	0.0000	43.6952	0.0000	1.0000	1.0000	1.0000
0xfb22...2305	0.0000	3.2204	0.0000	0.9601	0.9601	0.9601
0x6505...ffa5	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
0x7789...e051	0.0000	3.4428	0.0000	0.9680	0.9680	0.9680
0xf2c9...f2d5	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
0x8388...493d	0.0000	2.4052	0.0000	0.9098	0.9098	0.9098

Table 3: Case 3 (Apriori), Cluster 1 – full listing of all 9 noise wallets

Wallet	x_{graph}	x_{beh}	C_{graph}	C_{beh}	C_{uni}	$C(w)$
0x1a4b...067d	1.0000	9.1667	0.6321	0.9999	1.0000	1.0000
0x1fbe...508b	22.7500	0.0000	1.0000	0.0000	1.0000	1.0000
0xdc5b...ae1b	1.0000	5.0448	0.6321	0.9936	0.9976	0.9976
0xe5c0...a639	1.0000	9.1667	0.6321	0.9999	1.0000	1.0000
0x30f3...b2c7	1.0000	9.1667	0.6321	0.9999	1.0000	1.0000
0x92c2...81ea	1.0000	9.1667	0.6321	0.9999	1.0000	1.0000
0xfad0...321b	1.0000	9.1667	0.6321	0.9999	1.0000	1.0000
0xecb4...020b	0.0000	5.0000	0.0000	0.9933	0.9933	0.9933
0x0423...9958	0.0000	12.5000	0.0000	1.0000	1.0000	1.0000

Table 4: Case 3 (Apriori), Cluster 2 – full listing of all 8 noise wallets

Wallet	x_{graph}	x_{beh}	C_{graph}	C_{beh}	C_{uni}	$C(w)$
0x0210...6d5f	1.0000	10.0000	0.6321	1.0000	1.0000	1.0000
0x01c9...0bb0	28.2500	0.0000	1.0000	0.0000	1.0000	1.0000
0x6030...f9b2	0.0000	8.3333	0.0000	0.9998	0.9998	0.9998
0xacd9...3db6	0.0000	3.3610	0.0000	0.9653	0.9653	0.9653
0x6d04...b930	1.0000	8.3333	0.6321	0.9998	0.9999	0.9999
0xd4ae...8c1c	1.0000	10.0000	0.6321	1.0000	1.0000	1.0000
0x5af8...11f6	1.0000	8.3339	0.6321	0.9998	0.9999	0.9999
0x2961...b1c2	1.0000	3.5133	0.6321	0.9702	0.9890	0.9890

Derived from backward-path (incoming) influence rather than forward-path (outgoing) influence.

.2 Detailed Subgroup Cohesion for Different eps Values

Table 5: Subgroup Cohesion for eps = 0.2

Cluster	Subgroup	Cohesion
0	0	0.9062
0	1	0.8544
0	2	0.8600
0	3	0.8778
1	0	0.9887
1	1	0.8459
1	2	0.9255
1	3	0.8939
1	4	0.8095
2	0	0.8612
2	1	0.8754
3	0	0.8653
3	1	0.8560
3	2	0.8875
3	3	0.9468

Table 6: Subgroup Cohesion for eps = 0.15

Cluster	Subgroup	Cohesion
0	0	0.9016
0	1	0.9962
0	2	0.8754
0	3	0.9967
0	4	0.9460
0	5	0.9963
1	0	0.8536
1	1	1.0000
1	2	0.8939
1	3	0.9383
2	0	1.0000
2	1	0.9030
2	2	0.9468
2	3	1.0000
2	4	1.0000
2	5	0.8800
2	6	1.0000
3	0	0.9711
3	1	0.9206
3	2	1.0000
3	3	0.9797
3	4	1.0000
3	5	0.8908
3	6	1.0000
3	7	1.0000
3	8	1.0000
3	9	0.9513
3	10	1.0000

Table 7: Subgroup Cohesion for eps = 0.1

Cluster	Subgroup	Cohesion
0	0	0.9383
0	1	0.9358
0	2	0.9076
0	3	0.9160
0	4	1.0000
1	0	0.9030
1	1	1.0000
1	2	1.0000
1	3	1.0000
1	4	1.0000
1	5	1.0000
1	6	1.0000
1	7	1.0000
1	8	1.0000
2	0	0.9967
2	1	0.9243
2	2	0.9963
2	3	0.9962
2	4	0.9460
2	5	0.9965
3	0	0.8908
3	1	1.0000
3	2	1.0000
3	3	1.0000
3	4	0.9206
3	5	1.0000
3	6	0.9711
3	7	1.0000
3	8	0.9797
3	9	1.0000
3	10	1.0000

Table 8: Subgroup Cohesion for eps = 0.05

Cluster	Subgroup	Cohesion
0	0	0.9383
0	1	0.9358
0	2	0.9076
0	3	0.9160
0	4	1.0000
1	0	0.9030
1	1	1.0000
1	2	1.0000
1	3	1.0000
1	4	1.0000
1	5	1.0000
1	6	1.0000
1	7	1.0000
1	8	1.0000
2	0	0.9967
2	1	0.9243
2	2	0.9963
2	3	0.9962
2	4	0.9460
2	5	0.9965
3	0	0.8908
3	1	1.0000
3	2	1.0000
3	3	1.0000
3	4	0.9206
3	5	1.0000
3	6	0.9711
3	7	1.0000
3	8	0.9797
3	9	1.0000
3	10	1.0000